

Heteroskedastic Canonical Polyadic Tensor Decomposition

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Abstract

When minimizing the squared-error loss, the popular CP decomposition can be interpreted as parameter inference in a Gaussian model with a low-rank mean tensor and constant variance across the tensor entries. We introduce heteroskedastic-CP (HCP), which models entrywise variability with a non-constant, low-rank precision tensor, and develop an alternating block-coordinate ascent method to recover both the low-rank mean and precision tensors from noisy observations. Our procedure is computationally competitive, with the same leading-order factor-update complexity as CP-ALS. We demonstrate HCP on synthetic experiments and an EEG application.

1 Introduction

The canonical polyadic (CP) decomposition, also known as the CANDECOMP/PARAFAC decomposition, is a fundamental tool in tensor analysis [Hitchcock, 1927, Harshman, 1970, Carroll and Chang, 1970, Kolda and Bader, 2009] that serves as a higher-order generalization of the matrix singular value decomposition. Specifically, the CP decomposition approximates a tensor $\mathcal{X} \in \mathbb{R}^{N_1 \times N_2 \times \dots \times N_D}$ as a sum of R rank-one tensors:

$$\mathcal{X} \approx \sum_{r=1}^R \lambda_r \mathbf{b}_r^{(1)} \circ \mathbf{b}_r^{(2)} \circ \dots \circ \mathbf{b}_r^{(D)} = \llbracket \boldsymbol{\lambda}; \mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_D \rrbracket. \quad (1)$$

Here, $\circ$ denotes the outer product for $\mathbf{b}_r^{(d)}$ column vectors, and the equality is a notational switch to the Kruskal operator with factor matrices (see Section 1.2). CP decomposition can be used for interpretable factorization, dimensionality reduction, imputation, and compression, and has found a wide range of applications [Murphy et al., 2013, Jaffé et al., 2014, Cong et al., 2015, Williams et al., 2018, Kenny et al., 2005, Kolda and Bader, 2006, Opsahl and Panzarasa, 2009, Chen et al., 2002, Muti and Bourennane, 2005, Beylkin et al., 2009, Acar and Yener, 2009, Kolda and Bader, 2009].

As pointed out by Hong et al. [2020], the ℓ_2 optimization problem posed by the CP decomposition

$$\arg \min_{\boldsymbol{\lambda} \in \mathbb{R}^R, \mathbf{B}_1 \in \mathbb{R}^{N_1 \times R}, \dots, \mathbf{B}_D \in \mathbb{R}^{N_D \times R}} \|\mathcal{X} - \llbracket \boldsymbol{\lambda}; \mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_D \rrbracket\|_F^2 \quad (2)$$

is exactly the maximum likelihood estimation (MLE) problem for $\mathcal{B} := \llbracket \boldsymbol{\lambda}; \mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_D \rrbracket$ under the assumption that $\mathcal{X}$ follows an entrywise independent Gaussian model with low-rank



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mean $\mathcal{B}$, and constant variance σ^2 , i.e. $\mathcal{X}_j \sim \mathcal{N}(\mathcal{B}_j, \sigma^2)$, where $\mathbf{j} = (j_1, \dots, j_D)$ is a multi-index. In tensor datasets, though, the noise level is often not constant: some sensors may be noisier than others, or some experimental conditions measured with different precision. Ignoring this heteroskedasticity can bias the recovered factors toward high-variance regions of the data.

In this paper we propose the Heteroskedastic CP (HCP) Decomposition, which will both estimate the variance structure and adjust the mean decomposition accordingly. In this sense, HCP is a “two-parameter” extension of standard CP. Maintaining the MLE framework, our HCP model assumes that our data tensor $\mathcal{X} \in \mathbb{R}^{N_1 \times \dots \times N_D}$ is normally distributed, entrywise independently, i.e.

$$\mathcal{X}_j \sim \mathcal{N}(\mathcal{B}_j, \mathcal{S}_j^{-1}). \quad (3)$$

where both $\mathcal{B}$ and $\mathcal{S}$ are low-rank parameter tensors of the same dimensions as $\mathcal{X}$ (low-rank parametrization of $\mathcal{S}$ regularizes the otherwise intractable entrywise variance estimation). The joint log-likelihood (up to constants) is

$$\ell(\mathcal{B}, \mathcal{S} \mid \mathcal{X}) = \frac{1}{2} \left(\sum_j \log(\mathcal{S}_j) - \sum_j \mathcal{S}_j (\mathcal{X}_j - \mathcal{B}_j)^2 \right). \quad (4)$$

1.1 Summary of Contributions

We propose an alternating block coordinate ascent procedure for optimizing Equation (4) in Section 4. Despite the additional expressivity of HCP, this procedure is computationally efficient, with the cost per factor update dominated by the same term as in CP-ALS (Alternating Least Squares).

Our procedure relies on two novel subroutines that optimize Equation (4) with either the precision tensor or the mean tensor held fixed. These are called Weighted CP, presented in Section 2, and Gamma-rate CP, in Section 3.

The joint log-likelihood of Equation (4) is unbounded above, making the maximization problem degenerate. To see this, consider a single index $\mathbf{j}$ where an estimate $\widehat{\mathcal{B}}_j$ drives the residual $(\mathcal{X}_j - \widehat{\mathcal{B}}_j)$ to zero; then, the corresponding estimate $\widehat{\mathcal{S}}_j$, and the likelihood alongside it, will increase to ∞ . This degeneracy is analogous to finding the variance of a single number, and occurs, for instance, in Gaussian mixture models when a cluster is selected to contain a single point. We return to this issue in Section 4, and relate Equation (4) to a penalized likelihood that avoids the degeneracy altogether.

Synthetic experiments in Section 5 demonstrate the improved estimation capabilities of HCP under heteroskedastic noise and showcase the unique ability of our “two-parameter” decomposition to adapt to unknown or misspecified generating distributions. Finally, we apply HCP to an EEG dataset in Section 6, where estimating the variance provides additional insights unavailable to a practitioner relying on CP alone.

1.2 Notation & Preliminaries

In this section, we will provide a brief review of relevant tensor properties and discuss our notation. We provide a more detailed Tensor review, define each matrix product, and provide some necessary prerequisite propositions in Appendix A.

We denote vectors (first-order tensors) by bold lowercase letters (e.g., $\mathbf{a}$), matrices (second-order tensors) by bold uppercase letters (e.g., $\mathbf{A}$), and higher-order (order three or greater) tensors by Euler script letters (e.g., $\mathcal{X}$). The identity matrix of size $n \times n$ is $\mathbf{I}_n$, and the length- n vector of all ones as $\mathbf{1}_n$. The bracket notation $[n]$ denotes the set $\{1, \dots, n\}$. For a tensor $\mathcal{X} \in \mathbb{R}^{N_1 \times \dots \times N_D}$, the entry at multi-index $\mathbf{j} = (j_1, \dots, j_D)$, with $j_d \in [N_d]$, is written $\mathcal{X}_j$. The following shorthands will be used throughout: $N_{-d} = \prod_{c \neq d} N_c$, $N_+ = \sum_{d=1}^D N_d$, and $N_\times = \prod_{d=1}^D N_d$. We define $\text{dv}(\mathcal{X}) := \text{diag}(\text{vec}(\mathcal{X}))$, where $\text{diag}(\mathbf{x})$ is the diagonal matrix with

$\mathbf{x}$ as its diagonal, and $\text{vec}(\mathcal{X})$ denotes the vectorization. We write, $\mathcal{X}_{(d)} \in \mathbb{R}^{N_d \times N_{-d}}$ for mode- d matricization, and $\|\mathcal{X}\|_F = \|\text{vec}(\mathcal{X})\|_2$ denotes the Frobenius norm.

We denote the Kronecker product of $\mathbf{A} \in \mathbb{R}^{I \times J}$ and $\mathbf{B} \in \mathbb{R}^{K \times L}$ as $\mathbf{A} \otimes \mathbf{B} \in \mathbb{R}^{IK \times JL}$, and the Khatri–Rao product (“matching columnwise” Kronecker product) of $\mathbf{A} \in \mathbb{R}^{I \times K}$ and $\mathbf{B} \in \mathbb{R}^{J \times K}$ as $\mathbf{A} \odot \mathbf{B} \in \mathbb{R}^{IJ \times K}$. We denote the Hadamard (elementwise) product, division, and power by $\mathbf{A} * \mathbf{B}$, $\mathbf{A} \oslash \mathbf{B}$, and $\mathbf{A}^{*b}$ respectively. We may also use each matrix product over a range of matrices, e.g. $\odot_{d=1}^D \mathbf{A}_d = \mathbf{A}_1 \odot \cdots \odot \mathbf{A}_D$.

The *rank* of a tensor [Kolda and Bader, 2009] is the smallest integer R such that $\mathcal{X}$ can be written as a sum of R rank-one tensors. Similarly, the nonnegative rank [Lim and Comon, 2009], is the smallest integer R such that $\mathcal{X}$ can be written as a sum of R rank-one tensors, where each rank-one tensor has nonnegative factors. Note however, that in this article we will only work with data tensors with unknown rank (such as $\mathcal{X}$), and parameter tensors with known Kruskal form, such as $\mathcal{B} = \llbracket \boldsymbol{\lambda}; \mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_D \rrbracket$. Strictly speaking, if $\mathcal{B}$ has a rank- R Kruskal form then $\text{rank}(\mathcal{B}) \leq R$. Kruskal tensors have a scaling ambiguity on the factor columns because $\llbracket \boldsymbol{\lambda} * \boldsymbol{\gamma}; \mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_D \rrbracket = \llbracket \boldsymbol{\lambda}; \mathbf{B}_1 \text{diag}(\boldsymbol{\gamma}), \mathbf{B}_2, \dots, \mathbf{B}_D \rrbracket$ holds for any vector $\boldsymbol{\gamma}$. Therefore, throughout this paper we will assume that the matrix factors of Kruskal tensors have unit column norms, while $\boldsymbol{\lambda} \in \mathbb{R}_{\geq 0}^R$ contains the scaling factor. Strictly speaking this normalization is not required but it avoids roundoff errors that can occur in extreme cases [Ballard and Kolda, 2025]. When a normalization step is required in our procedures, we use the function `COLUMNNORMALIZE( $\tilde{\mathcal{B}}, p$ )` which returns a tuple $(\mathcal{B}, \boldsymbol{\lambda})$ such that $\mathcal{B}$ has unit- p norms and $\tilde{\mathcal{B}} = \mathcal{B} \text{diag}(\boldsymbol{\lambda})$ holds. We provide this function in the Appendix A.

1.3 Related Work

Leveraging the connection between maximum likelihood estimation and CP decomposition to motivate more sophisticated CP decomposition alternatives is not unique to our work. Chi and Kolda [2012] improve on previous non-negative tensor factorizations by explicitly modeling the sparse count tensor with the Poisson distribution rather than the Gaussian distribution, and generalized CP (GCP) decomposition Hong et al. [2020] demonstrates the advantage of their loss function flexibility through connections to different MLE problems. Heteroskedasticity, however, has not been as widely studied, particularly in CP decomposition.

For Tucker-parameterized models, recent works have proposed methods which are more adaptive to unknown variance profiles, including heteroskedasticity. For example, the subgaussian method discussed in Han et al. [2022] replaces the SVD initialization with HeteroskedasticPCA initialization, which allows them to consider non-identically distributed entries, and Agterberg and Zhang [2024] improve HOOI (higher-order orthogonal iteration) with more robust diagonal deletion initialization. To the best of our knowledge, our work is still the first tensor decomposition to explicitly estimate a heteroskedasticity tensor jointly with the low-rank mean, which can provide interpretability benefits, as we see in Section 6.

Heteroskedasticity has been much more extensively studied for matrix data [Hong et al., 2023, 2021, Zhang et al., 2022, Agterberg et al., 2022, Cai et al., 2021, Leeb and Romanov, 2021, Zhang and Mondelli, 2024, Su and Wu, 2025, Behne and Reeves, 2022]. We discuss our work’s relation to these methods in Appendix B.

2 Weighted CP Decomposition

In this section, we optimize Equation (4) when the precision tensor $\mathcal{S}$ is fixed. This subroutine amounts to solving a weighted CP decomposition problem.

Let $\mathcal{S} \in \mathbb{R}_{\geq 0}^{N_1 \times \dots \times N_D}$ be known “weights”, and consider the objective function

$$\ell(\mathcal{B} \mid \mathcal{X}, \mathcal{S}) = -\frac{1}{2} \sum_j \mathcal{S}_j (\mathcal{X}_j - \mathcal{B}_j)^2. \quad (5)$$

For entrywise positive $\mathcal{S}$, Equation (5) corresponds to the likelihood for our generative model (Equation (3)) when the precision tensor $\mathcal{S}$ is fixed and known. In addition, for binary weights, Equation (5) corresponds to the masked least-squares objective used in CP tensor completion, solved with a joint gradient approach called CP-WOPT by Acar et al. [2011a]. We explore the connection to tensor completion in Appendices G and H.

We can express our optimization problem as a weighted least squares problem via the following steps:

$$\begin{aligned}\ell(\mathcal{B} \mid \mathcal{X}, \mathcal{S}) &= -\frac{1}{2} \sum_j \mathcal{S}_j (\mathcal{X}_j - \mathcal{B}_j)^2 = -\frac{1}{2} \left\| \mathcal{S}_{(d)}^{*\frac{1}{2}} * (\mathcal{X}_{(d)} - \tilde{\mathcal{B}}_d \mathbf{\Pi}_d) \right\|_F^2 \\ &= -\frac{1}{2} \left\| \text{dv} \left((\mathcal{S}_{(d)}^{*\frac{1}{2}})^\top \right) \left(\text{vec}(\mathcal{X}_{(d)}^\top) - (\mathbf{I}_{N_d} \otimes \mathbf{\Pi}_d^\top) \text{vec}(\tilde{\mathcal{B}}_d^\top) \right) \right\|_2^2\end{aligned}\quad (6)$$

The first equality follows from Proposition 3, isolating a single factor matrix in the typical ALS fashion, with $\tilde{\mathcal{B}}_d := \mathcal{B}_d \text{diag}(\boldsymbol{\lambda}_{\mathcal{B}})$ and $\mathbf{\Pi}_d := (\odot_{c \neq d} \mathcal{B}_c)^\top \in \mathbb{R}^{R \times N_d}$. The second equality takes the transpose, vectorizes, and leverages Proposition 4 to convert into a typical least squares form. The transpose is taken so that the Gram matrix becomes block diagonal.

The weighted least-squares expression of Equation (6) has optima characterized by the normal equations

$$\mathbf{G}_d \text{vec}(\tilde{\mathcal{B}}_d^\top) = \text{vec} \left(\mathbf{\Pi}_d (\mathcal{S} * \mathcal{X})_{(d)}^\top \right), \quad (7)$$

where $\mathbf{G}_d$ is an $RN_d \times RN_d$ block-diagonal matrix whose (i_d, i_d) block, for $i_d \in [N_d]$, is

$$\mathbf{\Pi}_d \text{diag}((\mathcal{S}_{(d)})_{i_d,:}) \mathbf{\Pi}_d^\top \in \mathbb{R}^{R \times R}.$$

Because $\mathbf{G}_d$ is block diagonal, the linear system separates across the rows $(\mathcal{B}_d)_{i_d,:}$ of $\mathcal{B}_d$ and can be solved in parallel. As in standard CP-ALS, one can solve these equations in multiple ways depending on numerical conditioning [Ballard and Kolda, 2025]. In our implementation, we use the Moore–Penrose pseudoinverse.

To estimate $\mathcal{B} := \llbracket \boldsymbol{\lambda}_{\mathcal{B}}; \mathcal{B}_1, \mathcal{B}_2, \dots, \mathcal{B}_D \rrbracket$ we sequentially apply Equation (7) to each $d = 1, 2, \dots, D$ for $\leq i_{\mathcal{B}}$ iterations (or until convergence; see Appendix G for a discussion of convergence criterion). Algorithm 1 displays the full procedure, and we present a toy example in Appendix C.1.

When $\mathcal{S}$ is a dense tensor of weights, the complexity of Algorithm 1 is dominated by line 12, which has a cost of $O(R_{\mathcal{B}}^2 N_{\times})$ floating point operations for updating one factor matrix. If $\mathcal{S}$ is sparse (as in tensor completion), then this complexity drops to $O(R_{\mathcal{B}}^2 \text{nnz}(\mathcal{S}))$, where $\text{nnz}(\cdot)$ denotes the number of non-zeros. This decrease from $N_{\times}$ to $\text{nnz}(\mathcal{S})$ can be of several orders of magnitude. Furthermore, if the rank- $R_{\mathcal{S}}$ CP factorization $\mathcal{S} = \llbracket \boldsymbol{\lambda}_{\mathcal{S}}; \mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_D \rrbracket$ holds (as it is the case in our proposed HCP), then $\mathbf{G}_d$ of Equation (7) has the closed form

$$\mathbf{G}_d = \sum_{r=1}^{R_{\mathcal{S}}} (\boldsymbol{\lambda}_{\mathcal{S}})_r \text{diag}((\mathcal{S}_d)_{:,r}) \otimes \begin{pmatrix} * \\ c \neq d \end{pmatrix} \mathbf{H}_{c,r},$$

where $\mathbf{H}_{c,r} := \mathcal{B}_c^\top \text{diag}((\mathcal{S}_c)_{:,r}) \mathcal{B}_c \in \mathbb{R}^{R \times R}$. Forming this matrix has a complexity of only $O(R_{\mathcal{B}}^2 N_{\times} + R_{\mathcal{S}})$, which is considerably cheaper than $O(R_{\mathcal{B}}^2 N_{\times})$ or $O(R_{\mathcal{B}}^2 \text{nnz}(\mathcal{S}))$ (see Appendix F for complexity derivations). In this case, the computational reduction is so drastic that complexity is now dominated by the matricized tensor times Khatri–Rao product (MTTKRP) of line 10. This MTTKRP complexity is the same operation that dominates other CP estimation algorithms such as CP-ALS, CP-APR or CP-WOPT Kolda and Bader [2009], Chi and Kolda [2012], Hong et al. [2020]. So, in our implementation of Algorithm 1 we allow $\mathcal{S}$ to be provided as a factored tensor.

Algorithm 1 Weighted CP Decomposition

```

1: Input: data  $\mathcal{X} \in \mathbb{R}^{N_1 \times \dots \times N_D}$ 
2: Input: weights  $\mathcal{S} \in \mathbb{R}_{\geq 0}^{N_1 \times \dots \times N_D}$  given as dense, sparse, or factored (faster computation of
   line 12)
3: Input: desired CP rank  $R_{\mathcal{B}}$ 
4: Input: maximum number of iterations  $i_{\mathcal{B}}$ 
5: Initial Values:  $\mathbf{B}_2, \dots, \mathbf{B}_D$ , unit column  $\ell_2$  norms
6: function WCP( $\mathcal{X}, R_{\mathcal{B}}, \mathcal{S}, i_{\mathcal{B}}$ )
7:   while not converged and iterations  $\leq i_{\mathcal{B}}$  do
8:     for  $d = 1, \dots, D$  do
9:        $\mathbf{\Pi}_d \leftarrow (\odot_{c \neq d} \mathbf{B}_c)^\top$ 
10:       $\mathbf{Z}_d \leftarrow \mathbf{\Pi}_d (\mathcal{S} * \mathcal{X})_{(d)}^\top$  (MTTKRP)
11:      for  $i_d = 1, \dots, N_d$  do
12:         $\mathbf{G} \leftarrow \mathbf{\Pi}_d \text{dv}((\mathcal{S}_{(d)})_{i_d,:}) \mathbf{\Pi}_d^\top$ 
13:         $(\mathbf{B}_d)_{i_d,:} \leftarrow \text{solution } \mathbf{b} \text{ of } \mathbf{G}\mathbf{b} = \mathbf{Z}_{:,i_d}$ 
14:      end for
15:       $(\mathbf{B}_d, \lambda_{\mathcal{B}}) \leftarrow \text{COLUMNNORMALIZE}(\mathbf{B}_d, 2)$ 
16:    end for
17:  end while
18:  return CP factors:  $(\lambda_{\mathcal{B}}, \mathbf{B}_1, \dots, \mathbf{B}_D)$ 
19: end function

```

3 Gamma-Rate CP Decomposition

Next, we consider the model of Equation (3) when the mean tensor $\mathcal{B}$ is fixed, and derive an algorithm for estimating the positive factor matrices of $\mathcal{S}$, $\mathcal{S}_d$.

The log-likelihood, up to constants, is

$$\begin{aligned}
\ell(\mathcal{S} \mid \mathcal{X}, \mathcal{B}) &= \frac{1}{2} \left(\sum_j \log(\mathcal{S}_j) - \sum_j \mathcal{S}_j (\mathcal{X}_j - \mathcal{B}_j)^2 \right) \\
&= \frac{1}{2} \mathbf{1}_{N_d}^\top \left(\log(\mathcal{S}_{(d)}) - \mathcal{S}_{(d)} * (\mathcal{X} - \mathcal{B})_{(d)}^{*2} \right) \mathbf{1}_{N-d} \\
&= \frac{1}{2} \mathbf{1}_{N_d}^\top \left(\log(\tilde{\mathcal{S}}_d \mathbf{\Omega}_d) - (\tilde{\mathcal{S}}_d \mathbf{\Omega}_d) * \mathcal{R}_{(d)} \right) \mathbf{1}_{N-d},
\end{aligned} \tag{8}$$

where $\mathbf{\Omega}_d := (\odot_{c \neq d} \mathcal{S}_c)^\top \in \mathbb{R}_{>0}^{R \times N-d}$ (using Prop 3), $\tilde{\mathcal{S}}_d := \mathcal{S}_d \text{diag}(\lambda_{\mathcal{S}})$, and $\mathcal{R} = (\mathcal{X} - \mathcal{B})^{*2}$ is the squared residual. Because $\mathcal{S}_j \mathcal{R}_j \sim \chi_1^2 \equiv \text{Gamma}(\frac{1}{2}, \frac{1}{2})$, Equation (8) is the loglikelihood under

$$\mathcal{R}_j \sim \text{Gamma}\left(\frac{1}{2}, \text{rate} = \frac{1}{2} \llbracket \lambda_{\mathcal{S}}; \mathcal{S}_1, \dots, \mathcal{S}_D \rrbracket_j\right). \tag{9}$$

Hence, we refer to this subproblem as Gamma-rate CP decomposition, since we parametrize the rate (equivalently, the inverse mean), rather than the mean as in Section 3.3 of Hong et al. [2020].

Inspired by NNMF, PTF, and NNFT Lee and Seung [2000], Welling and Weber [2001], Chi and Kolda [2012], we develop a majorize-minimize update rule for $\tilde{\mathcal{S}}_d$. While a general-purpose constrained optimizer such as GCP could solve our problem, our result guarantees monotonic non-decrease without requiring a step size, and enforces nonnegativity by construction, rather than relying on box constraints. We defer the proof to Appendix E.

Theorem 1. *Let $\mathbf{A}, \mathbf{C} \in \mathbb{R}_{\geq 0}^{m \times n}$, $\mathbf{B} \in \mathbb{R}_{\geq 0}^{m \times p}$, and $\mathbf{\Omega} \in \mathbb{R}_{\geq 0}^{p \times n}$ such that $\mathbf{B}\mathbf{\Omega}$ and $\mathbf{C}\mathbf{\Omega}^\top$ are entry-wise positive. Then the function $f(\mathbf{B}) = \mathbf{1}_m^\top \{\mathbf{A} * \log(\mathbf{B}\mathbf{\Omega}) - \mathbf{C} * (\mathbf{B}\mathbf{\Omega})\} \mathbf{1}_n$ is non-decreasing*

under the multiplicative update

$$\mathbf{B} \leftarrow \mathbf{B} * \{[\mathbf{A} \oslash (\mathbf{B}\mathbf{\Omega})] \mathbf{\Omega}^\top\} \oslash (\mathbf{C}\mathbf{\Omega}^\top). \quad (10)$$

Applied to Equation (8), Theorem 1 translates (take $\mathbf{A}$ to be $\mathbf{1}_{N_d}\mathbf{1}_{N-d}^\top$, $\mathbf{B}$ to be $\tilde{\mathbf{S}}_d$, $\mathbf{\Omega}$ as $\mathbf{\Omega}_d$ and $\mathbf{C}$ as $\mathcal{R}_{(d)}^1$) to the following multiplicative update rule:

$$\tilde{\mathbf{S}}_d \leftarrow \tilde{\mathbf{S}}_d * \left((\tilde{\mathbf{S}}_d \mathbf{\Omega}_d)^{* -1} \mathbf{\Omega}_d^\top \right) \oslash (\mathcal{R}_{(d)} \mathbf{\Omega}_d^\top). \quad (11)$$

To estimate $\mathcal{S} := \llbracket \lambda_{\mathcal{S}}; \mathbf{S}_1, \mathbf{S}_2, \dots, \mathbf{S}_D \rrbracket$ we sequentially apply Equation (11) to each $d = 1, 2, \dots, D$ until convergence. Algorithm 2 displays the full procedure, and present a toy example in Appendix C.2. Note, in this algorithm we normalize with ℓ_1 rather than ℓ_2 norm.

Algorithm 2 Gamma-rate CP Decomposition

```

1: Input: Data:  $\mathcal{R} \in \mathbb{R}_{>0}^{N_1 \times N_2 \times \dots \times N_D}$ 
2: Input: Desired rank of results:  $R_{\mathcal{S}}$ 
3: Input: Maximum inner iterations:  $i_{\text{in}}$ 
4: Input: Maximum outer iterations  $i_{\mathcal{S}}$ 
5: Initial values:  $\lambda_{\mathcal{S}} \in \mathbb{R}_{>0}^{R_{\mathcal{S}}}$ ,  $(\mathbf{S}_1, \dots, \mathbf{S}_D)$ , with unit column  $\ell_1$  norms
6: function GAMMACP( $\mathcal{R}, R_{\mathcal{S}}, i_{\mathcal{S}}, i_{\text{in}}$ )
7:   while not converged and iterations  $\leq i_{\mathcal{S}}$  do
8:     for  $d = 1, \dots, D$  do
9:        $\mathbf{\Omega}_d \leftarrow (\odot_{c \neq d} \mathbf{S}_c)^\top$ 
10:       $\mathbf{D}_d \leftarrow \mathcal{R}_{(d)} \mathbf{\Omega}_d^\top$ 
11:       $\tilde{\mathbf{S}}_d \leftarrow \mathbf{S}_d \cdot \text{diag}(\lambda_{\mathcal{S}})$ 
12:      for  $i = 1, \dots, i_{\text{in}}$  do
13:         $\tilde{\mathbf{S}}_d \leftarrow \tilde{\mathbf{S}}_d * \left( (\tilde{\mathbf{S}}_d \mathbf{\Omega}_d)^{* -1} \mathbf{\Omega}_d^\top \right) \oslash \mathbf{D}_d$ 
14:      end for
15:       $\{\mathbf{S}_d, \lambda_{\mathcal{S}}\} \leftarrow \text{COLUMNNORMALIZE}(\tilde{\mathbf{S}}_d, 1)$ 
16:    end for
17:  end while
18:  return CP Factors:  $\{\lambda_{\mathcal{S}}, \mathbf{S}_1, \dots, \mathbf{S}_D\}$ 
19: end function

```

4 Heteroskedastic CP Decomposition

Now, we provide the full procedure in Algorithm 3 for jointly estimating a low-rank mean tensor, $\mathcal{B}$, and a low-rank precision tensor $\mathcal{S}$, from an observed noisy tensor $\mathcal{X}$ that we assume follows the generative model (3). As discussed in Section 1, the idea is to alternate between estimating the mean tensor $\mathcal{B}$ with Algorithm 1, assuming $\mathcal{S}$ is fixed and known, and estimating $\mathcal{S}$ with Algorithm 2, assuming $\mathcal{B}$ is fixed and known.

For initialization, we use standard CP decomposition to initialize $\mathbf{B}_d$, and Algorithm 2 to initialize $\mathbf{S}_d$. In our experiments (Sections 5, 6), ranks will be either known or tailored to the application. Appendix G discusses data-driven rank selection and other modeling choices, including convergence criteria, maximum iterations, and settings for Algorithms 1 and 2.

The time complexity of Algorithm 3 is

$$O(i_{\text{tot}} [i_{\mathcal{B}} D N_{\times} R_{\mathcal{B}} + i_{\mathcal{S}} i_{\text{in}} D N_{\times} R_{\mathcal{S}}]), \quad (12)$$

¹ $\mathbf{C}\mathbf{\Omega}^\top = \mathcal{R}_{(d)}(\mathbf{\Omega}_d)^\top$ will have an entry of 0 if an entire length N_{-d} row of the matricized squared residual tensor, $\mathcal{R}_{(d)}$, is exactly 0. Such exact row-wise interpolation is unlikely when $\mathcal{X}$ is noisy, and the constraint discussed in Section 4.3 prevents this altogether.

Algorithm 3 Heteroskedastic CP Decomposition

```

1: Input: Data  $\mathcal{X} \in \mathbb{R}^{N_1 \times N_2 \times \dots \times N_D}$ 
2: Input: Desired ranks of results:  $R_{\mathcal{B}}, R_{\mathcal{S}} > 0$ 
3: Input: Submethod iterations:  $i_{\mathcal{B}}, i_{\mathcal{S}}, i_{\text{in}} > 0$ 
4: Input: Maximum iterations:  $i_{\text{tot}} > 0$ 
5: Initial values:  $(\mathbf{B}_2, \dots, \mathbf{B}_D)$  as in Alg. 1
6: Initial values:  $(\boldsymbol{\lambda}_{\mathcal{S}}, \mathbf{S}_1, \dots, \mathbf{S}_D)$  as in Alg. 2
7: while not converged and iterations  $\leq i_{\text{tot}}$  do
8:    $\{\boldsymbol{\lambda}_{\mathcal{B}}, \mathbf{B}_1, \dots, \mathbf{B}_N\} \leftarrow \text{WCP}(\mathcal{X}, R_{\mathcal{B}}, \mathcal{S}, i_{\mathcal{B}})$ 
9:    $\mathcal{R} \leftarrow (\mathcal{X} - \llbracket \boldsymbol{\lambda}_{\mathcal{B}}; \mathbf{B}_1, \dots, \mathbf{B}_N \rrbracket)^{*2}$ 
10:   $\{\boldsymbol{\lambda}_{\mathcal{S}}, \mathbf{S}_1, \dots, \mathbf{S}_N\} \leftarrow \text{GAMMACP}(\mathcal{R}, R_{\mathcal{S}}, i_{\mathcal{S}}, i_{\text{in}})$ 
11: end while
12: Output:  $\mathbf{B}_d \in \mathbb{R}^{N_d \times R}$  for  $d = 1, \dots, D$ 
13: Output:  $\mathbf{S}_d \in \mathbb{R}_{>0}^{N_d \times R}$  for  $d = 1, \dots, D$ 

```

where the first and second terms correspond to Algorithms 1 and 2, respectively. For reference, a sweep of CP-ALS has a cost of $O(DN_{\times}R_{\mathcal{B}})$ which is the same cost as our Algorithm 1. In most practical applications of our algorithm (small $R_{\mathcal{S}}$ or partial heteroskedasticity, which we discuss next) the cost of Algorithm 2 will be dominated by Algorithm 1, making our HCP algorithm and CP-ALS have similar performance. Deeper discussion and full derivations are deferred to the Appendix F.

As mentioned earlier, our objective function can be unbounded when our residual tensor $\mathcal{R}$ has entries equal to or approaching 0. Next, we discuss ways to avoid this degeneracy.

4.1 Rank-One Precision Tensor

When the precision rank is one, the MM update (Equation (11)) reduces to

$$\mathbf{s}_d \leftarrow N_{-d} (\mathcal{R}_{(d)} \boldsymbol{\Omega}_d)^{*^{-1}},$$

where $\mathbf{s}_d$ and $\boldsymbol{\Omega}_d = \mathbf{s}_D \odot \dots \odot \mathbf{s}_{d+1} \odot \mathbf{s}_{d-1} \odot \dots \odot \mathbf{s}_1 \in \mathbb{R}^{N-d}$ are now *vectors*. For fixed other factors, or along sequences where $\boldsymbol{\Omega}_d$ is bounded below, an entry of this weighted residual contraction only tends to zero when the entire corresponding residual row tends to zero, which is unlikely in most practical scenarios where $\mathcal{B}$ serves as a low-rank approximation of the noisy tensor $\mathcal{X}$. Hence, we have found that setting $R_{\mathcal{S}} = 1$ tends to lead to a stable optimization problem.

4.2 Partial Heteroskedasticity

Our second alternative is “partial heteroskedasticity”, i.e., allowing the precision tensor $\mathcal{S}$ to vary along certain modes but not others. This assumption is common in applications where certain modes correspond to the dimension of a process, and other modes to samples of the process, as in MANOVA [Johnson and Wichern, 2007], factor analysis [Anderson and Rubin, 1956], or heteroskedastic PCA Hong et al. [2023, 2021], Zhang et al. [2022].

To simplify notation (we can always permute the tensor modes), we will assume the first $P \leq D$ modes of $\mathcal{S}$ are heteroskedastic, while the final Q modes are homoskedastic.

This means that $\mathcal{B} = \llbracket \boldsymbol{\lambda}_{\mathcal{B}}; \mathbf{B}_1, \dots, \mathbf{B}_D \rrbracket$ as usual, while $\mathcal{S}$ is written,

$$\mathcal{S} = \llbracket \boldsymbol{\lambda}_{\mathcal{S}}; \mathbf{S}_1, \dots, \mathbf{S}_P \rrbracket \circ \mathbf{1}_{I_{P+1}} \circ \dots \circ \mathbf{1}_{I_D} = \llbracket \boldsymbol{\lambda}_{\mathcal{S}}; \mathbf{S}_1, \dots, \mathbf{S}_P, \mathbf{1}_{I_{P+1}} \mathbf{1}_{R_{\mathcal{S}}}^{\top}, \dots, \mathbf{1}_{I_D} \mathbf{1}_{R_{\mathcal{S}}}^{\top} \rrbracket. \quad (13)$$

Implementing this restriction is simple. The only change in Algorithm 3 is in line 9, where

$\mathcal{R}$ is replaced with its average over the homoskedastic modes: for $\mathbf{i} \in [N_1] \times \dots \times [N_P]$

$$\mathcal{R}_{\mathbf{i}} = \frac{1}{\prod_{j=1}^Q N_{P+j}} \sum_{\mathbf{j} \in [N_{P+1}] \times \dots \times [N_D]} (\mathcal{X}_{(\mathbf{i}, \mathbf{j})} - \mathcal{B}_{(\mathbf{i}, \mathbf{j})})^2. \quad (14)$$

We also need to change the way the subroutines are initialized. The term $\mathcal{S}$ in line 8 is now based on the second equality of Equation (13). Similarly, line 10 is now initialized only on the first P modes.

When leveraging this restriction, the method is far less likely to suffer from the degeneracy discussed; even with $Q = 1$, $\ell(\mathcal{B}, \mathcal{S} | \mathcal{X}) \rightarrow \infty$ would require small residuals for an entire fiber. Observing the unbounded likelihood is still possible, though, particularly when $N_{P+1}, \dots, N_D$ are small and $\mathcal{S}$ has high rank.

4.3 Constrained Residual Tensor

We can avoid $l(\mathcal{S} | \mathcal{X}, \mathcal{B}) \rightarrow \infty$ (and prevent divisions by zero) altogether by replacing $\mathcal{R}$ with $\mathcal{R} + \delta$ in the $\mathcal{S}$ update, for the scalar $\delta > 0$. This approach changes the loglikelihood from Equation (8) to

$$\ell_{\delta}(\mathcal{S} | \mathcal{X}, \mathcal{B}) := \frac{1}{2} \mathbf{1}_{N_d}^{\top} \left[\log(\mathbf{S}_d \mathbf{\Omega}_d) - (\mathcal{R}_{(d)} + \delta) * \mathbf{S}_d \mathbf{\Omega}_d \right] \mathbf{1}_{N-d}. \quad (15)$$

The following theorem justifies optimizing ℓ_{δ} as penalized estimation of ℓ .

Theorem 2. *Consider the scalar $\delta > 0$. The functions $\ell_{\delta}(\mathcal{S} | \mathcal{X}, \mathcal{B})$ (Equation (15)) and $l(\mathcal{S} | \mathcal{X}, \mathcal{B})$ (Equation (8)) are related as*

$$\ell_{\delta}(\mathcal{S} | \mathcal{X}, \mathcal{B}) = l(\mathcal{S} | \mathcal{X}, \mathcal{B}) - \frac{\delta}{2} \|\mathcal{S}\|_1.$$

We defer the proof to the appendix, and note that Theorem 2 holds for $l(\mathcal{B}, \mathcal{S} | \mathcal{X})$ and $\ell_{\delta}(\mathcal{B}, \mathcal{S} | \mathcal{X})$ as well. Implementation is straightforward; the only change in Algorithm 3 is in line 9, where $\mathcal{R}$ now gets added δ , so $\mathcal{R} \leftarrow \mathcal{R} + \delta$. The $\mathcal{B}$ update still solves $l(\mathcal{B} | \mathcal{X}, \mathcal{S})$ since the penalty only depends on $\mathcal{S}$.

Unless specified otherwise, we set δ to be $\sqrt{\text{Machine-Precision}} := \epsilon$.

5 Experiments

5.1 Rotating Butterfly

Our first experiment demonstrates the improvement of HCP over existing methods in the presence of heteroskedastic noise. We model a 200×200 binary matrix representing a pixelated butterfly image taken from the MPEG-7 and available in the `fdasrvf` R package [Tucker, 2026], where the third mode contains approximately $\pi/8$ -radian pixelized rotations of the image over a full rotation (so $\mathcal{B} \in \mathbb{R}^{200 \times 200 \times 17}$).

Then, representing the common situation where the noise is largest where the signal is, we generate our precision tensor to have small entries in the center, and constant along the third mode (which corresponds to rotation). The precision frontal slices have maximum value of two in the corners, and decay exponentially toward the center with a rate λ , which “controls” the level of heteroskedasticity ($\lambda = 0$ represents constant entrywise variance). The precision tensor $\mathcal{S}$ is then standardized to have an average variance of 0.5, ensuring a constant SNR between values of λ . Figure 1 demonstrates a slice of $\mathcal{S}$ with $\lambda \in \{1, 2, 4\}$.

We aim to recover a low-rank $\hat{\mathcal{B}}$ tensor from $\mathcal{X}_j \sim \mathcal{N}(\mathcal{B}_j, \mathcal{S}_j^{-1})$, where $\mathcal{S}$ is generated with different levels of heteroskedasticity. We compare HCP with CP and the Poisson loss in GCP

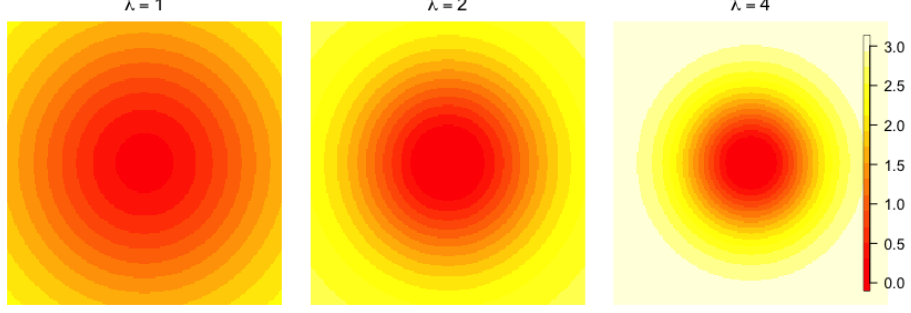


Figure 1: Frontal slice of $\mathcal{S}$ following heteroskedastic variance construction, for different values of λ .

Hong et al. [2020], denoted PGCP (GCP implementation details are in Appendix I). Since the Poisson distribution has equal mean and variance, PGCP is a sensible competitor in this scenario with high variance where the signal is largest. We set $R = 16$ for CP and PGCP, which was chosen such that the recovery in the $\lambda = 0$ case (constant variance) was strong. For HCP, we let $R_{\mathcal{B}} = 16$, but only set $R_{\mathcal{S}} = 3$, to limit the number of extra parameters HCP will have.

Figure 2 displays the relative error in terms of Frobenius norm, $\frac{\|\hat{\mathcal{B}} - \mathcal{B}\|_F}{\|\mathcal{B}\|_F}$, of the $\hat{\mathcal{B}}$ estimates for HCP and CP, for $\lambda \in \{0, 0.5, 1, 2, 3, 4, 6\}$, and we demonstrate the visual recovery for a single slice in Figure 3. As we can see, when there is no heteroskedasticity ($\lambda = 0$) or low heteroskedasticity, the methods perform roughly the same, but as λ grows CP and PGCP degrade significantly, while HCP does not.

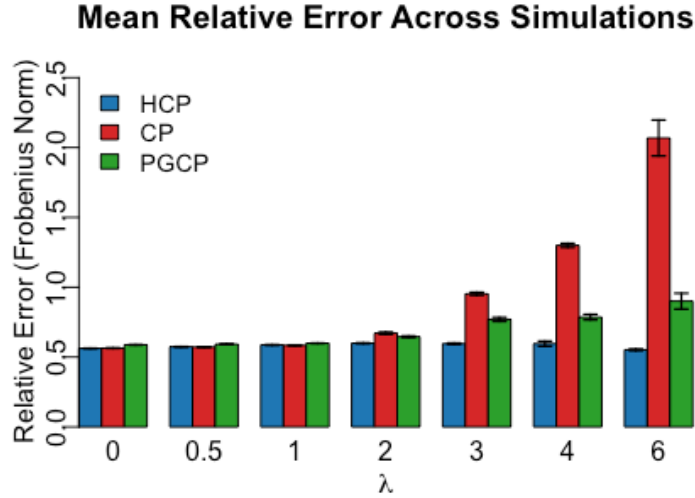


Figure 2: Comparison of relative errors in estimates of $\mathcal{B}$ at varying heteroskedasticity levels.

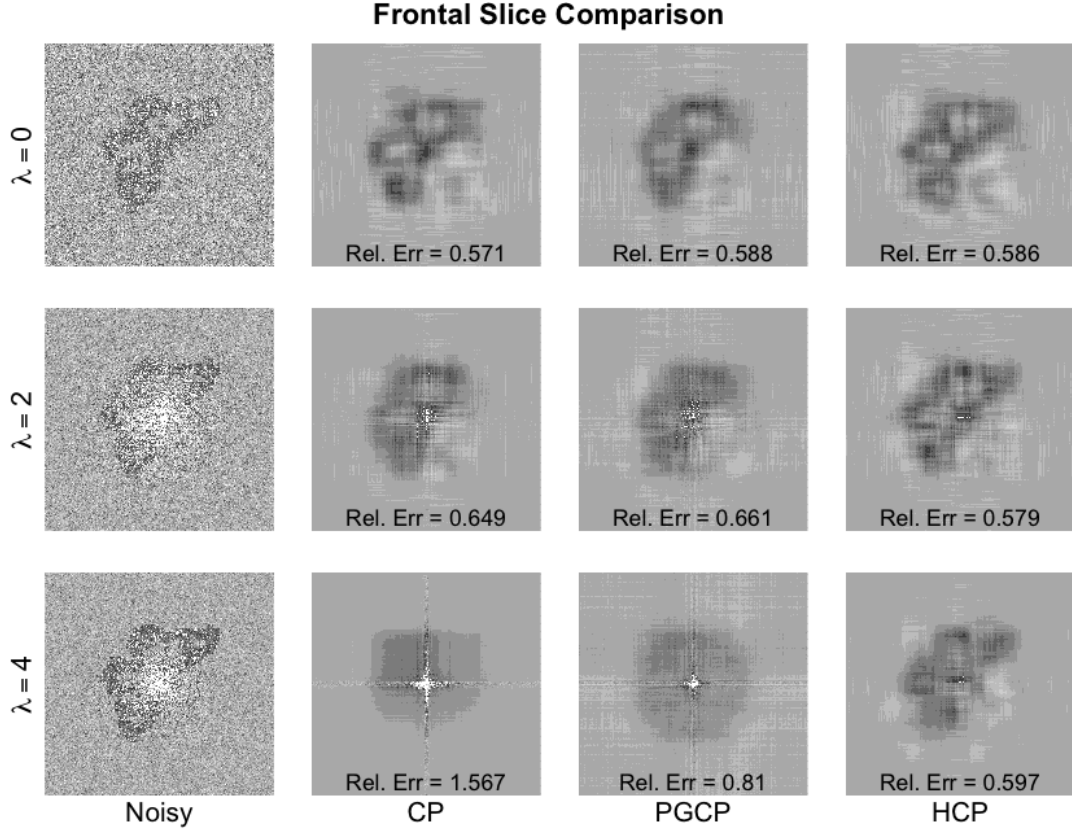


Figure 3: Visualization of 9th slice of the noisy observed tensor $\mathcal{X}$ and the CP, PGCP, and HCP estimates $\hat{\mathcal{B}}$, for varying heteroskedasticity levels.

5.2 Misspecified Generating Distribution

For our second experiment, we explore another advantage of HCP; by estimating both the conditional mean tensor and a second moment or variance-related tensor, HCP can be more safely employed when the exact noise model is not known.

We construct a simulation in which the true mean tensor $\mathcal{B} = \llbracket \lambda_{\mathcal{B}}; \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3 \rrbracket \in \mathbb{R}^{40 \times 32 \times 48}$ has low CP rank $R = 3$ with $1/R$ of the entries in each column of $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3$ generated from $\text{Unif}(10, 20)$, and the rest from $\text{Unif}(1, 2)$. The noisy observations are then generated from four distributions with different forms of heteroskedasticity:

$$X_i | \mu_i \sim \begin{cases} \text{Gaussian}(\mu_i, \sigma^2) & \text{Var}(X_i | \mu_i) \propto 1 \\ \text{Poisson}(\mu_i) & \text{Var}(X_i | \mu_i) \propto \mu_i \\ \text{Gamma}(\alpha, \text{rate} = \frac{\alpha}{\mu_i}) & \text{Var}(X_i | \mu_i) \propto \mu_i^2 \\ \text{Inv-Gaussian}(\mu_i, \lambda) & \text{Var}(X_i | \mu_i) \propto \mu_i^3 \end{cases},$$

where i indexes tensor entries and μ_i denotes the corresponding entry of the true mean tensor $\mathcal{B}$. We calibrated the free dispersion parameters $\sigma^2, \alpha, \lambda$ so that the average conditional variance was the same across data-generating mechanisms, using the Poisson model, which has no additional dispersion parameter, as the reference.

Figure 4 summarizes the relative errors in $\hat{\mathcal{B}}$ over 50 simulations, comparing the mean estimates from HCP, CP, and GCP decompositions [Hong et al., 2020] using Poisson, Gamma, and Inv-Gaussian losses (GCP implementation details are in Appendix I). As expected, the best fit comes from the correct distributional loss, but these methods can each struggle under different distributions. In contrast, HCP performs strongly in all cases, because even though

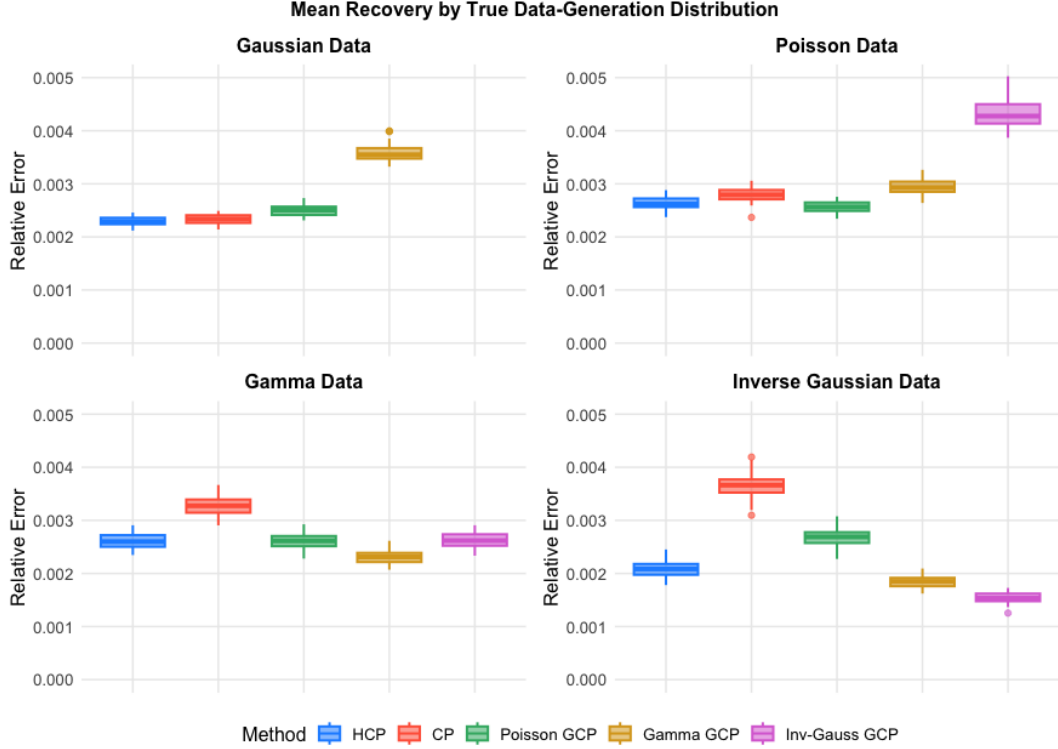


Figure 4: Panels correspond to a different true observation model, and boxplots show the $\hat{\mathcal{B}}$ recovery in relative Frobenius error over replicates for HCP, CP, and GCP methods using Poisson, Gamma, and inverse Gaussian losses.

our method has an inherent Gaussian assumption, it is flexible in its first two moments, and hence is more flexible and adapts to model misspecification.

5.3 Parameter Recovery Benchmark

Finally, we conduct a simulation study across varying tensor sizes, $m_{\text{mult}} \cdot (10 \times 8 \times 12)$, with $m_{\text{mult}} \in \{1.5, 2, 3, 6\}$, where we know the true low-rank tensors $\mathcal{B}$ and $\mathcal{S}$. We generate a partially heteroskedastic $\mathcal{S}$ that is homoskedastic along the third mode, and we estimate it by leveraging Section 4.2 accordingly. To analyze the effect of δ in Section 4.3, we use both $\delta = 0.001$ and $\delta = \sqrt{\text{Machine-Precision}} = \epsilon$, labeled HCP and HCP0, respectively. In Appendix D we provide more details on the simulation, present results for fully heteroskedastic precision, and analyze the effect of rank misspecification. We measure performance with relative errors, and for the mean recovery we will compare our method against CP decomposition. For the recovery of the precision, we are not aware of a directly comparable baseline.

Figure 5 provides our results as boxplots over 50 simulations. The mean recovery consistently outperforms CP decomposition, and there is almost no difference in the mean or precision recovery between HCP and HCP0, particularly for the larger tensor sizes. The sufficiency of a small constraint motivates our choice of $\delta = \epsilon$ as the default.

Some takeaways from the rank misspecification results in Appendix D are also worth noting, since in practical applications of HCP the true ranks will be unknown. First, it is generally more detrimental to underestimate the rank than it is to overestimate the rank, consistent with standard model selection discussions for CP [Bro and Kiers, 2003, Kolda and Bader, 2009]. Secondly, we have key validation that for even the worst selections of $R_{\mathcal{B}}$ or $R_{\mathcal{S}}$ we examined, the mean estimate $\hat{\mathcal{B}}$ from HCP will outperform its standard CP counterpart.

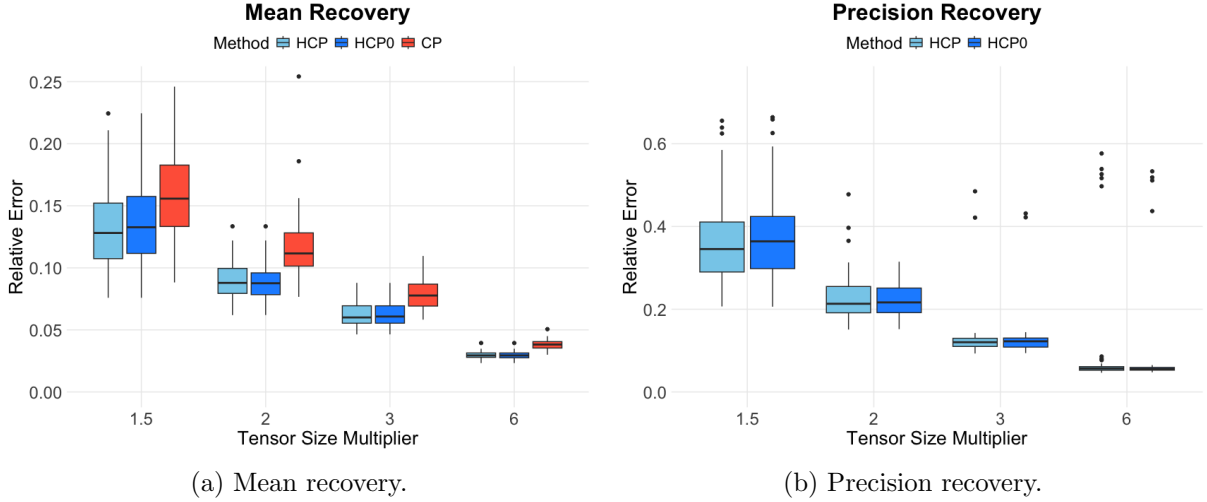


Figure 5: Simulation results comparing recovery performance across tensor sizes. The left panel shows relative error in recovery of $\mathcal{B}$, while the right panel shows relative error in recovery of $\mathcal{S}$.

6 EEG Data Analysis

In this section, we apply HCP to electroencephalogram (EEG) data, motivated by prior work using CP decompositions to recover interpretable factors from multi-channel EEG tensors [Cong et al., 2015, Acar et al., 2011b].

For our EEG dataset, we use the publicly available EEG Motor Movement/Imagery Dataset from PhysioNet [Goldberger et al., 2000, Schalk et al., 2009], collected using the BCI2000 system [Schalk et al., 2004]. This dataset contains 64-channel (i.e. electrode locations on the scalp) EEG recordings from subjects performing motor movement and imagery 2-minute tasks. We consider 3 “runs”, with 15 measurements each, of the first subject performing task 1, where a target appears on the left or the right side of the screen, and the subject opens and closes their corresponding fist. Following the preprocessing steps of Acar et al. [2011b] (with full details in Appendix J), we compute a continuous wavelet time-frequency transform for each channel and measurement, then vectorize the frequency-by-time representation into a 6099-dimensional feature vector. Hence, we obtain an observed tensor $\mathcal{X} \in \mathbb{R}^{64 \times 6099 \times 45}$, whose modes correspond to the 64 EEG channels, 6099 time-frequency coefficients, and 45 measurements.

We will analyze the low-rank recovery $\hat{\mathcal{B}}$ of $\mathcal{X}$, computed using $R_{\mathcal{B}} = 3$, matching the rank used in Acar et al. [2011b]. For HCP, we set $R_{\mathcal{S}} = 1$, and assume $\mathcal{S}$ is partially heteroskedastic, with constant variance along the measurement mode. Figure 6 displays the estimated log-variance by time-frequency location, where we observe elevated average variance at lower frequencies. This matches a physical interpretation for EEG data and validates HCP as a viable alternative: low-frequency components are typically more variable due to the well-known $1/f$ -like spectral structure of EEG [Pritchard, 1992], and they are also more susceptible to slow drift, motion artifacts, and ocular contamination [Jung et al., 2000, Delorme and Makeig, 2004].

The estimated mean factors for HCP and CP are presented in Figure 7. For both methods, the second component captures similar channel and time frequency structure that is stable across measurements, and the first component appears to capture a low-frequency pattern that differs between measurement runs, though the measurement loadings are much more random in CP than in HCP. The third component displays the most notable difference between the methods: HCP captures a clear distinction in a channel loading between runs, while CP appears to only capture an inverse of the low-frequency pattern in its first component. In Appendix J, Figure 16, we examine a rank-4 fit for CP, where the additional component still only seems to model this low-frequency pattern.

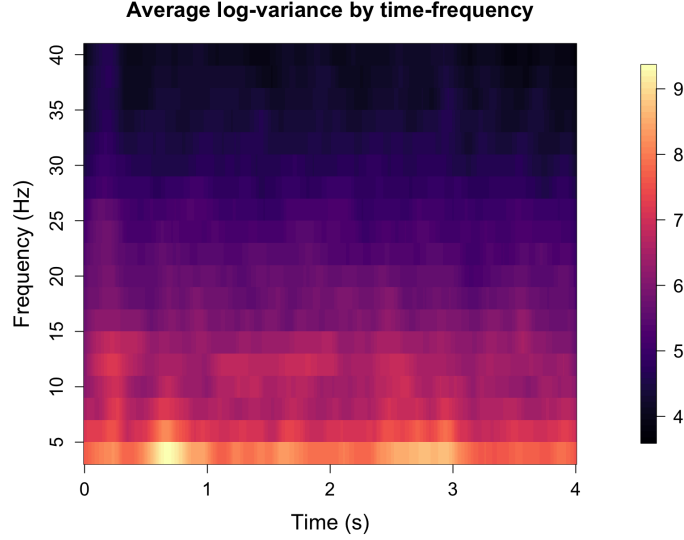


Figure 6: Estimated log-variance from HCP for the EEG tensor, averaged over channels and runs. The recovered variance is largest at lower frequencies.

Coupled with our estimate of the variance structure in Figure 6, it seems likely that CP decomposition is treating the low-frequency variability as stable mean signal, while HCP, by modeling the heteroskedasticity separately, produces interpretable factors less dominated by this high-variance region.

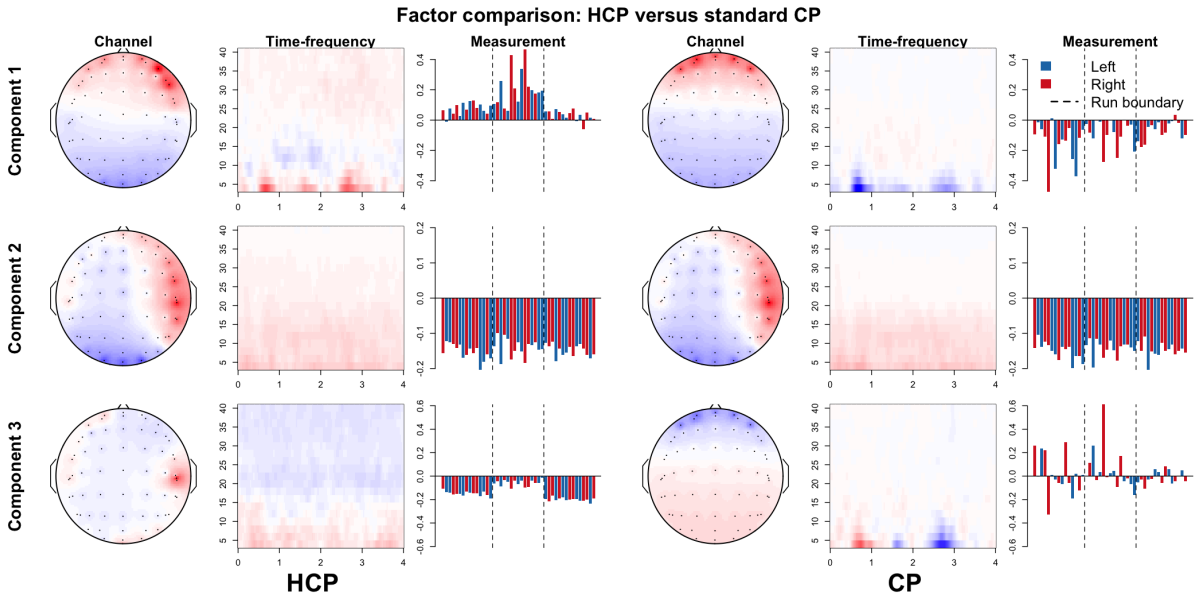


Figure 7: Visualization of the factor matrices recovered from the EEG tensor by CP and HCP with rows corresponding to components. Within each method, columns show the channel/scalp loadings, time-frequency loadings, and measurement loadings. Dashed vertical lines in measurement columns indicate recording-run boundaries.

7 Discussion

In this paper, we introduced Heteroskedastic CP (HCP) Decomposition, an extension of CP decomposition to an entrywise independent Gaussian model with unknown, low-rank precision tensor. We jointly estimate the mean and precision by alternating between the weighted CP and Gamma-rate CP subproblems, deriving block-coordinate updates for each, and address the degeneracy of the objective with a residual modification that corresponds to a penalized likelihood approach. Across synthetic experiments, we see that HCP is substantially more stable than CP or GCP decomposition under increasing heteroskedasticity, and we demonstrate the advantages of our “two-parameter” decomposition in handling model misspecification and providing interpretability benefits.

Addressing the assumptions of our model provides many clear avenues for future work. The entrywise independence could be replaced with a covariance from the tensor-multivariate-normal distribution Manceur and Dutilleul [2013], the Gaussian generating distribution could be replaced with one allowing heavy tails, such as Student’s t distribution, taking inspiration from Auddy and Yuan [2022], or other low-rank models (e.g. Tucker, Tensor Train) could be considered. Other two parameter distributions may also be of interest. For example, the non-negative tensor factorizations developed by Chi and Kolda [2012] would likely not be adaptive to over-dispersed regimes, mirroring CP’s degeneracy under heteroskedasticity.

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A A Short Primer on Tensor Operations

In this section, we provide a brief review of tensor properties and definitions we use in the main manuscript. We also provide two key propositions regarding these matrix products. We will use the N -way tensor $\mathcal{X} \in \mathbb{R}^{N_1 \times \dots \times N_D}$.

Fixing all indices of $\mathcal{X}$ except one yields a *fiber*, the higher-order analogue of a matrix row/column Kolda [2006]. Fixing all indices except two yields a *slice*, i.e., a two-dimensional section of the tensor. The *mode- d matricization* $\mathcal{X}_{(d)}$ (see Kolda [2006] for the general case), arranges the mode- d fibers as the columns of a matrix, so $\mathcal{X}_{(d)} \in \mathbb{R}^{N_d \times \prod_{m \neq d} N_m}$.

The tensor $\mathcal{X}$ is *rank-one* if it can be written as the outer product of N vectors, $\mathcal{X} = \mathbf{b}^{(1)} \circ \mathbf{b}^{(2)} \circ \dots \circ \mathbf{b}^{(d)}$, meaning $x_{i_1 \dots i_d} = \mathbf{b}_{i_1}^{(1)} \dots \mathbf{b}_{i_d}^{(d)}$. Tensor rank, denoted as $\text{rank}(\mathcal{X})$, is defined as the smallest integer R such that $\mathcal{X}$ can be written as a sum of R rank-one tensors Kolda [2006]. When referring to non-negative tensors such as $\mathcal{S}$, we will often use the non-negative rank [Lim and Comon, 2009], denoted $\text{rank}_+(\mathcal{S})$, and defined as the smallest integer R such that $\mathcal{S}$ can be written as a sum of R rank-one tensors, where each rank-one tensor has nonnegative factors. Note that it does not generally hold that $\text{rank}(\mathcal{S}) = \text{rank}_+(\mathcal{S})$.

When a CP-rank decomposition is known, we can express the tensor in terms of its factor matrices using the *Kruskal operator*. Specifically,

$$\mathcal{B} = \sum_{r=1}^R \lambda_r \mathbf{b}_r^{(1)} \circ \mathbf{b}_r^{(2)} \circ \dots \circ \mathbf{b}_r^{(D)} := \llbracket \boldsymbol{\lambda}; \mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_D \rrbracket, \quad (16)$$

where $\mathbf{b}_r^{(d)} \in \mathbb{R}^{N_d}$ are the factor column vectors for mode d , making up each rank-one component, and $\mathbf{B}_d = [\mathbf{b}_1^{(d)}, \mathbf{b}_2^{(d)}, \dots, \mathbf{b}_R^{(d)}] \in \mathbb{R}^{N_d \times R}$ is the factor matrix for mode d .

As mentioned in Section 1.2, Kruskal tensors have an inherent scaling ambiguity, hence our decision to write the scaling vector $\boldsymbol{\lambda}$ explicitly, and use factor matrices with unit column norms. We use the following function in Algorithms 1, 2, and 3 to normalize a factor matrix and return the scaling vector.

Algorithm 4 COLUMNNORMALIZE

```

1: Input: Matrix  $\mathbf{B} \in \mathbb{R}^{N \times R}$ , norm  $p \geq 1$ 
2: for  $r = 1, \dots, R$  do
3:    $\lambda_r \leftarrow \|\mathbf{B}_{:,r}\|_p$ 
4:   if  $\lambda_r > 0$  then
5:      $\mathbf{B}_{:,r} \leftarrow \mathbf{B}_{:,r} / \lambda_r$ 
6:   end if
7: end for
8: return  $(\mathbf{B}, \boldsymbol{\lambda})$ 
```

The Kronecker product of matrices $\mathbf{A} \in \mathbb{R}^{I \times J}$ and $\mathbf{B} \in \mathbb{R}^{K \times L}$ is denoted by $\mathbf{A} \otimes \mathbf{B}$. The result is a matrix of size $(IK) \times (JL)$ and defined by

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} \mathbf{A}_{11}\mathbf{B} & \mathbf{A}_{12}\mathbf{B} & \dots & \mathbf{A}_{1J}\mathbf{B} \\ \mathbf{A}_{21}\mathbf{B} & \mathbf{A}_{22}\mathbf{B} & \dots & \mathbf{A}_{2J}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{I1}\mathbf{B} & \mathbf{A}_{I2}\mathbf{B} & \dots & \mathbf{A}_{IJ}\mathbf{B} \end{bmatrix}.$$

The Khatri–Rao product is the “matching columnwise” Kronecker product. Given matrices $\mathbf{A} \in \mathbb{R}^{I \times K}$ and $\mathbf{B} \in \mathbb{R}^{J \times K}$, their Khatri–Rao product is denoted by $\mathbf{A} \odot \mathbf{B}$. The result is a matrix of size $(IJ) \times K$ defined by

$$\mathbf{A} \odot \mathbf{B} = [\mathbf{a}_1 \otimes \mathbf{b}_1 \quad \mathbf{a}_2 \otimes \mathbf{b}_2 \quad \dots \quad \mathbf{a}_K \otimes \mathbf{b}_K],$$

where $\mathbf{a}_k$ and $\mathbf{b}_k$ denote the k th columns of $\mathbf{A}$ and $\mathbf{B}$, respectively. If $\mathbf{a}$ and $\mathbf{b}$ are vectors, then the Khatri–Rao and Kronecker products are identical, i.e., $\mathbf{a} \otimes \mathbf{b} = \mathbf{a} \odot \mathbf{b}$.

Finally, we provide some key propositions of various matrix products we utilize for solving the likelihood equations in Sections 2 and 3. See Kolda [2006] and Petersen and Pedersen [2012] for a more comprehensive review of these matrix and tensor operations.

Proposition 3. (*Proposition 5.3(b) in Kolda [2006]*) *The mode- d matricization of Kruskal tensors:*

$$\llbracket \boldsymbol{\lambda}; \mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_D \rrbracket_{(d)} = \mathbf{X}_d \text{diag}(\boldsymbol{\lambda}) (\mathbf{X}_D \odot \dots \odot \mathbf{X}_{d+1} \odot \mathbf{X}_{d-1} \odot \dots \odot \mathbf{X}_1)^\top$$

Proposition 4. (*Equation (520) in Petersen and Pedersen [2012]*) $\text{vec}(\mathbf{A}\mathbf{X}\mathbf{B}^\top) = (\mathbf{B} \otimes \mathbf{A})\text{vec}(\mathbf{X})$.

B Heteroskedastic PCA

As mentioned in the main manuscript, Heteroskedasticity has been much more extensively studied for matrix principal component analysis (PCA) [Hong et al., 2023, 2021, Zhang et al., 2022] or for (low-rank) matrix estimation / matrix denoising [Agterberg et al., 2022, Cai et al., 2021, Leeb and Romanov, 2021, Zhang and Mondelli, 2024, Su and Wu, 2025, Behne and Reeves, 2022].

Modern heteroskedastic PCA formulations leverage what we call “partial heteroskedasticity” (defined in Section 4.2), where only some directions have heteroskedastic data. Zhang et al. [2022], Zhou and Chen [2025] model row-wise heteroskedasticity, which they call “within-sample” heteroskedasticity, as a sample is a column in the observation matrix. In a contrasting regime, Hong et al. [2021, 2016, 2018, 2023], model heteroskedasticity in PCA “between-samples” or column-wise, there are distinct noise groups and group memberships known a priori. Among these works, Hong et al. [2021] is closest to our setting in that estimating noise variances, rather than only obtaining robustness to heteroskedasticity, is a primary goal. Although its likelihood has an analogous degeneracy, the Gaussian prior on the low-dimensional coefficients regularizes the low-rank fit, so the variance update is not based on residuals from a freely optimized deterministic reconstruction as in HCP, and the degeneracy only occurs only when a sample lies exactly in the learned low-dimensional subspace. As in the rank-one case for HCP discussed in Section 4.1, such exact fits are unlikely for noisy data.

For matrix estimation/denoising, row-wise heteroskedasticity is popular [Agterberg et al., 2022, Cai et al., 2021, Su and Wu, 2025, Leeb and Romanov, 2021], though Behne and Reeves [2022] considers block-dependent signal strengths across pre-specified groups, for blocks that can be both row and column dependent, and Zhang and Mondelli [2024] considers a more general structured-noise model of the form $\mathbf{W} = \Xi^{1/2} \mathbf{Z} \Sigma^{1/2}$, which allows heterogeneity and correlations along both matrix dimensions. This final variance structure can be viewed as the order-2 (matrix) specialization of the Kronecker-separable covariance structure used for Tensor-on-Tensor Regression Llosa-Vite and Maitra [2023], and corresponds to a special case of HCP (Model 3) when the precision tensor $\mathcal{S}$ is rank-one. In particular, if the matrix noise model is written as $\mathbf{W} = \Xi^{1/2} \mathbf{Z} \Sigma^{1/2}$, with $\mathbf{Z}_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, $\Xi = \text{diag}(\xi_1, \dots, \xi_d)$, and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_d)$, then $\mathbf{W}_{ij} \sim \mathcal{N}(0, \mathbf{S}_{ij}^{-1})$, for $\mathbf{S}_{ij} = \xi_i^{-1} \sigma_j^{-1}$. This comparison primarily highlights the divergence, though, of Model 3 from the heteroskedastic matrix estimation literature: in HCP, the noise is generated entrywise independently, and the structured variability is encoded through the low-rank form of the precision tensor $\mathcal{S}$, rather than through an explicit covariance tensor.

C Toy Examples

C.1 Algorithm 1 Toy Example

For a toy demo of Algorithm 1, we consider low-rank $\mathcal{B} = \llbracket \lambda_{\mathcal{B}}; \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3 \rrbracket$ with $\mathbf{B}_1 \in \mathbb{R}^{10 \times 3}$, $\mathbf{B}_2 \in \mathbb{R}^{8 \times 3}$, $\mathbf{B}_3 \in \mathbb{R}^{12 \times 3}$, each generated entrywise randomly from the standard normal distribution.

Per the setting of this paper, we observe $\mathcal{X}_j \sim \mathcal{N}(\mathcal{B}_j, \mathcal{S}_j^{-1})$. $\mathcal{S}$ is taken to be fixed and known, with the corresponding variance tensor constructed to increase linearly along the first mode and vary quadratically along the third mode. In particular,

$$\mathcal{S}_{ijk} = \frac{1}{(0.5i)(0.25((k-3)^2 + 1))}, \quad i \in [I_1] \ j \in [I_2], \ k \in [I_3].$$

Figure 8 plots the first horizontal slice and the first lateral slice for $\mathcal{X}$, the true mean $\mathcal{B}$, and the recovered $\hat{\mathcal{B}}$ from Algorithm 1, which we see is a good estimate. The fourth column provides the recovered mean from CP decomposition, for comparison, which we see performs far worse. While this is not a fair comparison – with the known variance tensor, HCP simply has access to more information than CP – we get confirmation that Algorithm 1 can incorporate heteroskedasticity knowledge to improve our estimate of $\mathcal{B}$.

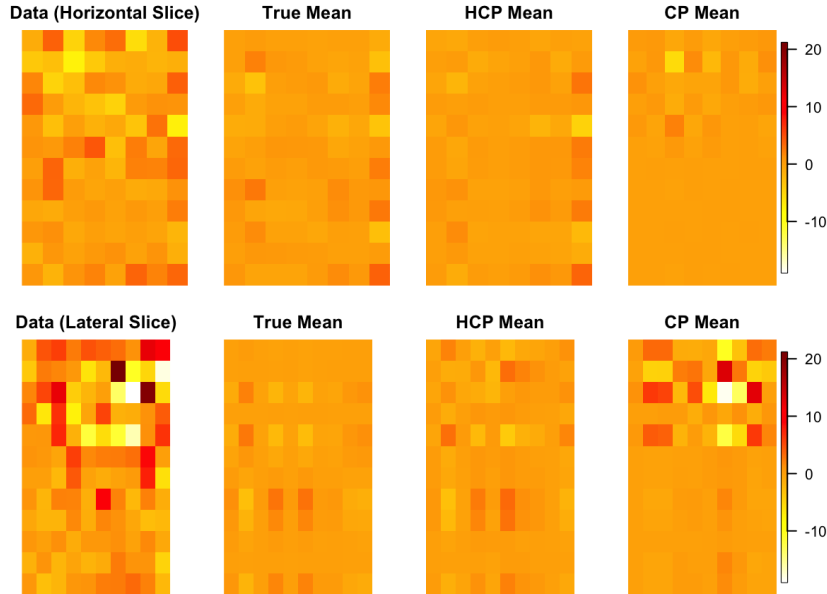


Figure 8: Results for a toy 3 mode tensor ($10 \times 8 \times 12$) with $R = 3$ and known $\mathcal{S}$. The first row displays the first horizontal slice, and the second, the first lateral slice, for the observed data, the true mean, and the estimated mean.

C.2 Algorithm 2 Toy Example

Here, we reverse the toy example given in Appendix C.1, setting $\mathcal{S} = \llbracket \lambda_{\mathcal{S}}; \mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3 \rrbracket$, with $\mathbf{S}_1 \in \mathbb{R}_+^{10 \times 3}$, $\mathbf{S}_2 \in \mathbb{R}_+^{8 \times 3}$, and $\mathbf{S}_3 \in \mathbb{R}_+^{12 \times 3}$. We generate the factor entries by randomly selecting approximately $1/R$ of the entries to be generated from $\text{Unif}(1, 2)$, and generate the remaining entries from $\text{Unif}(0.1, 0.2)$. This more complicated structure is necessary since the precision factor matrices are positive [Chi and Kolda, 2012]; if we used a single distribution for all entries, their common positive mean can reinforce an approximately rank-one structure in $\mathcal{S}$. For $\mathcal{B}$, we simply set it to zero, since only the residuals enter the precision update.

Figure 9 plots the first horizontal and lateral slices for the observed residuals $\mathcal{R}$, the true variance $\mathcal{S}^{*-1}$, and the recovered $\hat{\mathcal{S}}^{*-1}$ from Algorithm 2. Like the subroutine for estimating

the mean, given the precision, we see that this subroutine for estimating the precision, given the mean, has strong recovery. Without any competitors, though, we can't provide a comparison as we do in Appendix C.1.

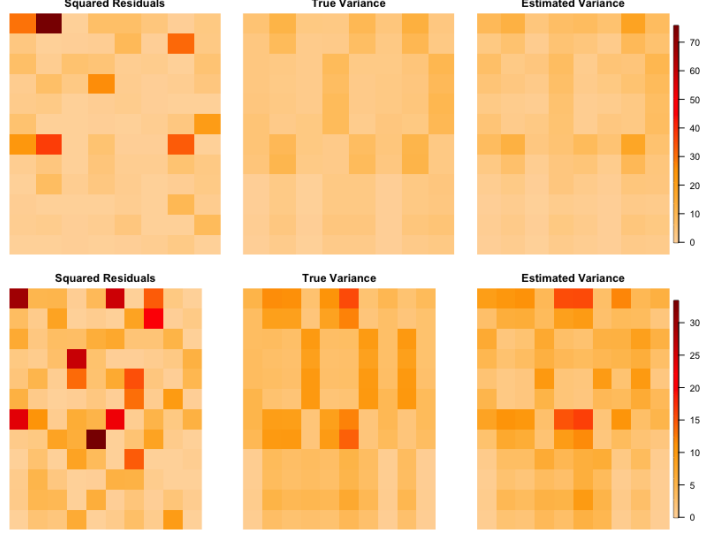


Figure 9: Results for a 3 mode tensor ($10 \times 8 \times 12$) with $R = 3$ and known $\mathcal{B} = 0$. The first row displays the first horizontal slice, and the second, the first lateral slice, for the squared residuals, the true variance, and the estimated variance.

C.3 Algorithm 3 Toy Example

Now, with strategies in place to alleviate the problems from the unbounded likelihood, we move on to a toy example, combining the examples of Appendix C.1 and C.2. Specifically, both $\mathcal{B}$ and $\mathcal{S}$ are rank 3, and the factor matrices are generated as in Sections C.1 and C.2, respectively. We also assume $\mathcal{S}$ is constant along the third mode, so that we may leverage the techniques in Section 4.2.

We set the constraint (as in Section 4.3) to $\delta = 0.001$. We plot the first frontal slices of the observed data, true mean and variance, and estimated mean and variance, in Figure 10, where we see that the recovery is strong for both parameters. We also provide the estimate for the mean from CP decomposition, for comparison, which does appear to be visually worse than HCP. Sections 5.3 and Appendix D expand this Toy Example into a robust simulation study, and compares with CP decomposition more thoroughly.

D Simulation Study

In this section, we provide more details regarding the simulation study in Section 5.3, as well as provide additional results for a fully heteroskedastic precision tensor. In addition, we analyze the effect of rank misspecification.

We will scale up the toy example conducted in Appendix C.3 to varying tensor sizes and assess the recovery over many simulations. Since the true low-rank tensors $\mathcal{B}$ and $\mathcal{S}$ are known, we use $R_{\mathcal{B}} = R_{\mathcal{S}} = 3$ and measure performance with Relative Errors $\frac{\|\mathcal{B} - \hat{\mathcal{B}}\|_F}{\|\mathcal{B}\|_F}$ and $\frac{\|\mathcal{S} - \hat{\mathcal{S}}\|_F}{\|\mathcal{S}\|_F}$. $\mathcal{X}$ will be size $m_{\text{mult}} \cdot (10 \times 8 \times 12)$, with $m_{\text{mult}} \in \{1.5, 2, 3, 6\}$. We will present here the results both for partial and fully heteroskedastic $\mathcal{S}$ (so $Q = 1$ and $Q = 0$, respectively, in the notation of Section 4.2), repeating the results from Section 5.3, where only partially heteroskedastic $\mathcal{S}$ is considered, for easier comparison. For our constraint we use $\delta = 0.001$ and $\delta = \sqrt{\text{Machine-Precision}}$, labeled HCP and HCP0, respectively. We will compare our method against CP decomposition for the mean recovery. For the recovery of the precision, there are no other competitors.

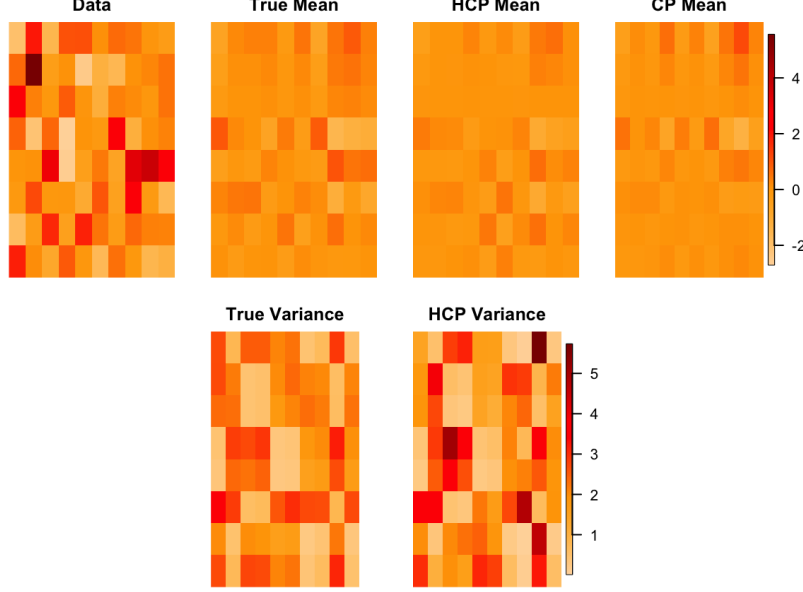
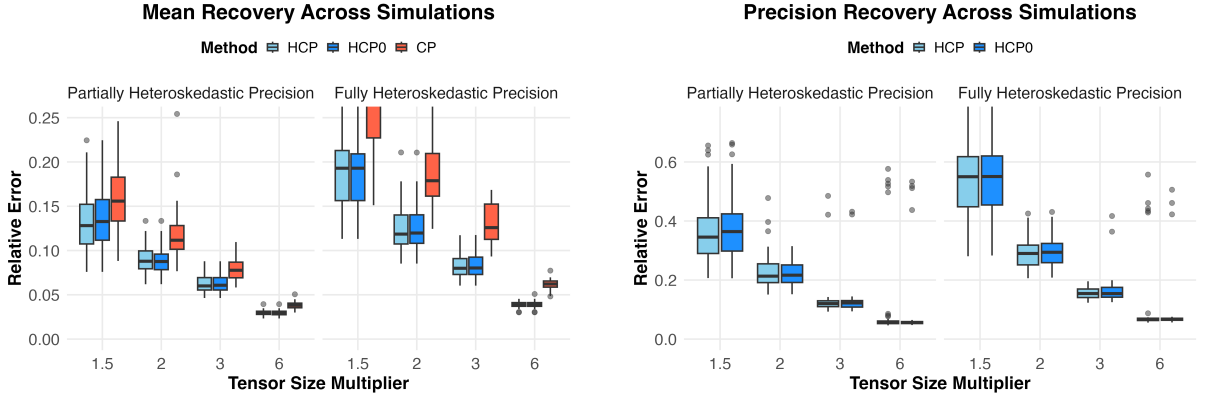


Figure 10: Results for a 3 mode tensor ($10 \times 8 \times 12$) with $R = 3$. For the first frontal slice we show the data, true mean tensor, and estimated mean tensor in the first row, and true vs learned variance tensors in the second row.

Figure 11 provides our results as boxplots over 50 simulations. As we can see from Figure 11a, our methods, regardless of the constraint or the exploitation of partial heteroskedasticity, outperform CP decomposition. For the recovery of the precision tensor, we have no competitors to compare against, but, as expected, our relative errors decrease with the size of the tensor. In addition, there is almost no difference in the mean or precision recovery between HCP and HCP0, particularly for the larger tensor sizes. This motivates our choice of $\delta = \sqrt{\text{Machine-Precision}}$ as the default, since in many cases this small constraint is sufficient. Finally, as expected, estimation when $\mathcal{S}$ is fully heteroskedastic is a more difficult task.



(a) Relative error for mean recovery across simulations.

(b) Relative error for precision recovery across simulations.

Figure 11: Simulation results comparing recovery performance across tensor sizes and heteroskedasticity structures. The left panel shows relative error in recovery of $\mathcal{B}$, while the right panel shows relative error in recovery of $\mathcal{S}$.

Next, we will intentionally fit models with ranks different from the truth ($R = 3$) in order to assess whether performance degrades gracefully under rank misspecification. Rather than considering a full grid over mean and precision ranks, we vary one rank at a time while holding

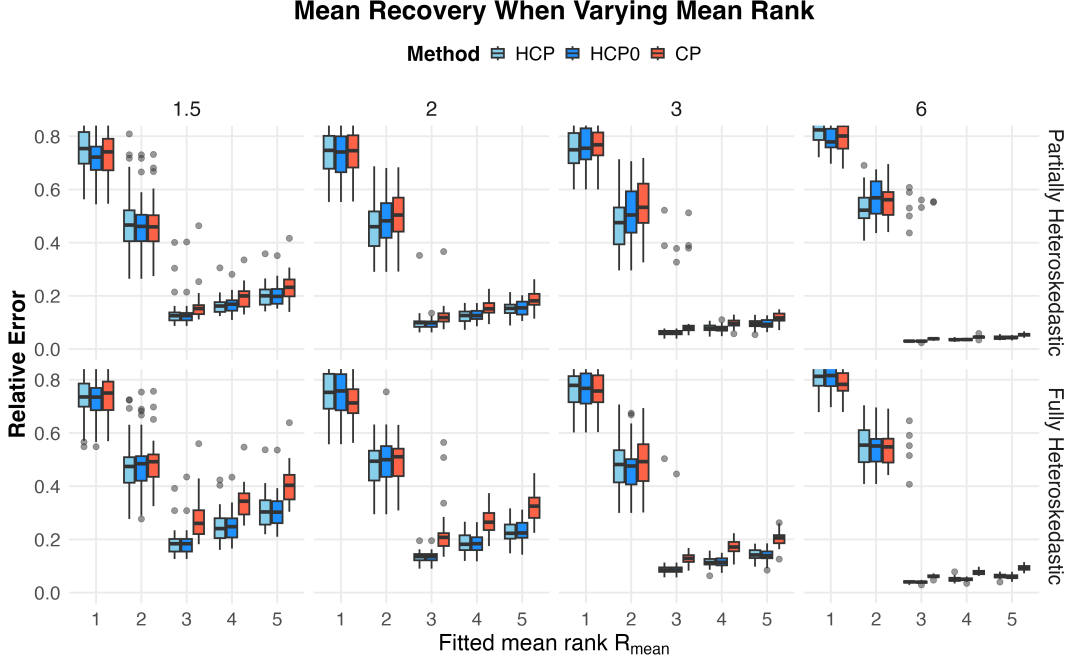


Figure 12: Mean recovery when varying the fitted mean rank R_B , with the precision rank fixed at its true value $R_S = 3$. The true mean rank is $R_B = 3$. HCP consistently outperforms CP decomposition, with the worst performance for both methods occurring when the fitted mean rank is too small.

the other fixed at its true value. Specifically, we consider two rank-robustness experiments:

$$R_B \in \{1, 2, 3, 4, 5\}, \quad R_S = 3,$$

and

$$R_B = 3, \quad R_S \in \{1, 2, 3, 4, 5\}.$$

This design isolates the effect of misspecifying the rank of the mean model from the effect of misspecifying the rank of the precision model. Ordinary CP decomposition is included as a baseline when studying mean recovery; for the varying mean-ranks, CP is fit with the corresponding mean rank. Performance is again measured using relative Frobenius error.

Figures 12–15 summarize the results. Figure 12 shows mean recovery when the fitted mean rank is varied while the precision rank is fixed at its true value. Figure 13 shows precision recovery under this same mean-rank variation, illustrating how misspecification of the mean model can also affect estimation of the heteroskedastic structure through the residuals. Figure 14 shows mean recovery when the precision rank is varied while the mean rank is fixed at its true value, assessing whether mean estimation is sensitive to misspecification of the variance model. Finally, Figure 15 shows precision recovery when the fitted precision rank is varied. The labels “Partially Heteroskedastic” and “Fully Heteroskedastic” indicate whether the precision tensor is constrained to be constant along one mode or allowed to vary along all modes.

As we can see, it is generally more detrimental to underestimate the rank than it is to overestimate the rank, consistent with standard model selection discussions for CP [Bro and Kiers, 2003, Kolda and Bader, 2009]. However, for small tensors, the precision recovery does heavily deteriorate when the fitted rank is too large, indicating a likely concern with overfitting in these cases.

Importantly, HCP still outperforms CP decomposition in the mean recovery (Figures 12 and 14) in every case. Particularly relevant is Figure 14, where we are analyzing the recovery of $\hat{\mathcal{B}}$ where $R_B = 3$ and R_S varies; the mean recovery will improve upon correct rank specification, but

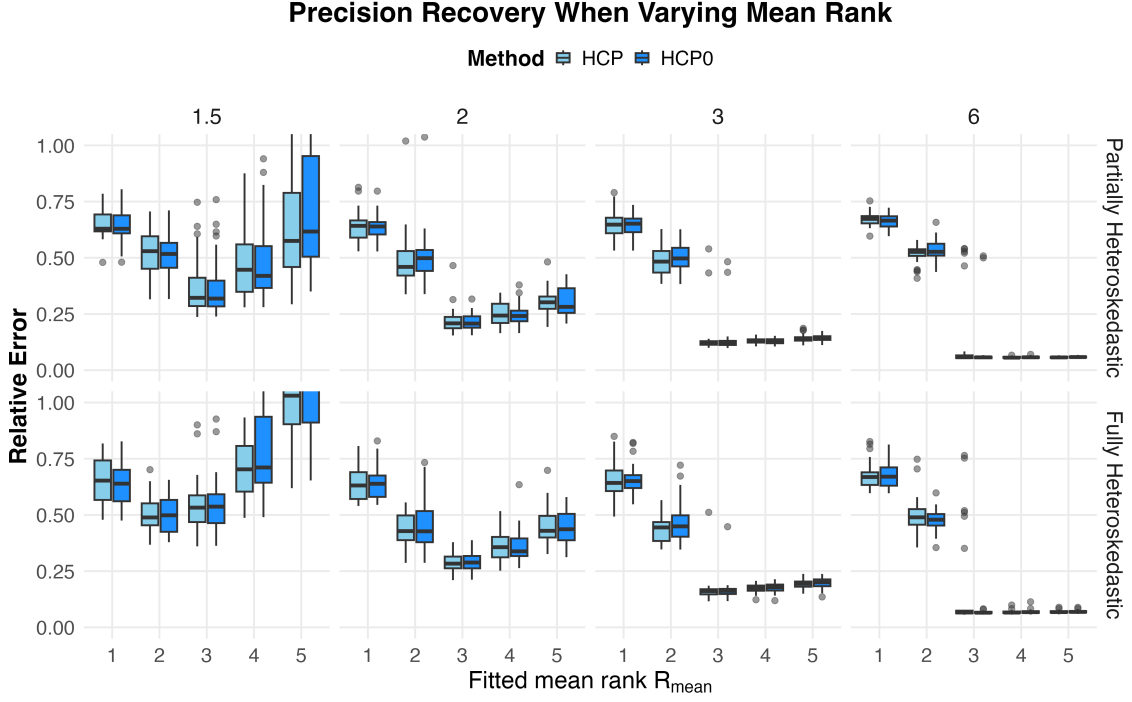


Figure 13: Precision recovery when varying the fitted mean rank $R_{\mathcal{B}}$, with the precision rank fixed at its true value $R_{\mathcal{S}} = 3$. This plot shows how misspecification of the mean rank affects estimation of the precision tensor.

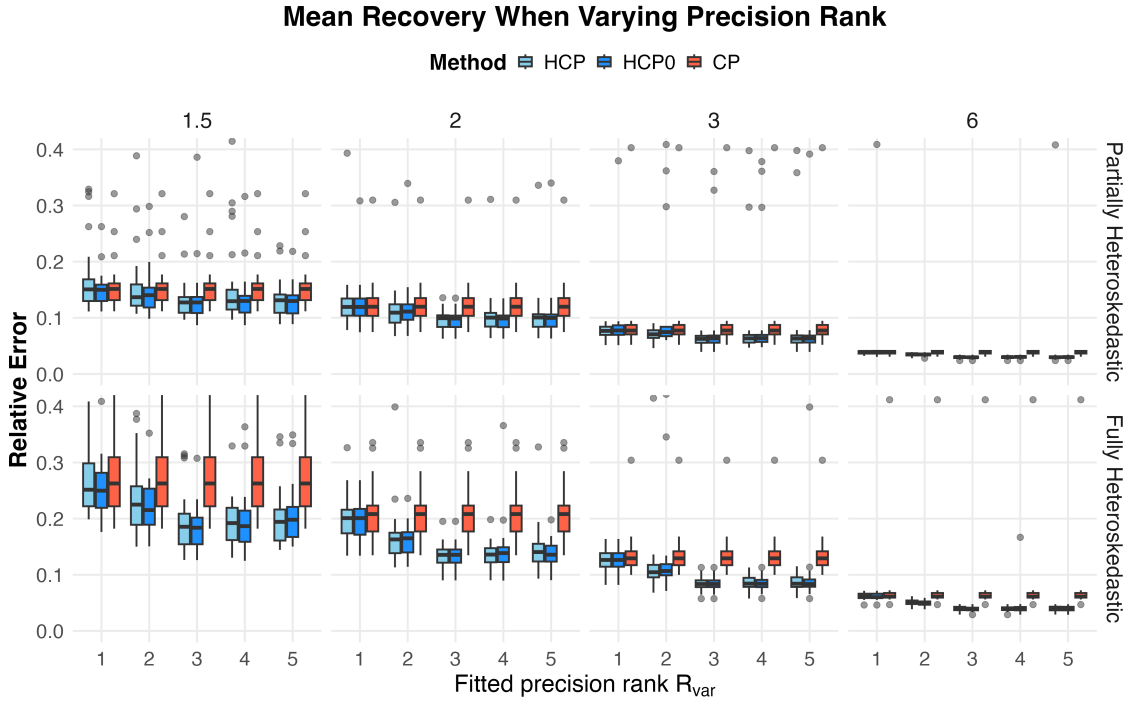


Figure 14: Mean recovery when varying the fitted precision rank $R_{\mathcal{S}}$, with the mean rank fixed at its true value $R_{\mathcal{B}} = 3$. Importantly, even for the worst fitted precision ranks, HCP will outperform CP in mean recovery.

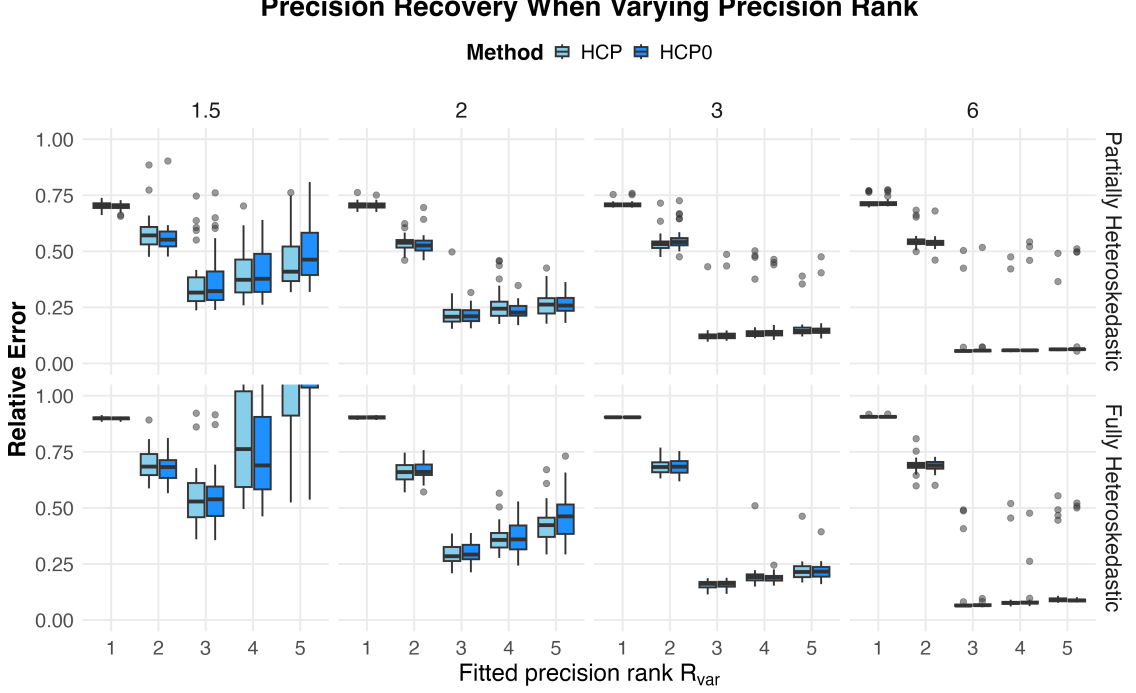


Figure 15: Precision recovery when varying the fitted precision rank R_S , with the mean rank fixed at its true value $R_B = 3$. The true precision rank is $R_S = 3$. Unlike for the mean, if the tensor is small, the fitted rank being too large can be more detrimental than being too small.

even in the most extreme case of $R_S = 1$, the estimate of $\hat{\mathcal{B}}$ from HCP outperforms that of CP decomposition. Combined, Section 5.2 and these results provide evidence that, in the presence of heteroskedastic noise, the mean estimate benefits from incorporating noise information, even if characterization of the noise structure is incorrect.

E Proof of Theorems 1 and 2

In this section, we prove Theorem 1.

Theorem 1 Let $\mathbf{A}, \mathbf{C} \in \mathbb{R}_{\geq 0}^{m \times n}$, $\mathbf{B} \in \mathbb{R}_{\geq 0}^{m \times p}$, and $\mathbf{\Omega} \in \mathbb{R}_{\geq 0}^{p \times n}$ such that $\mathbf{B}\mathbf{\Omega}$ and $\mathbf{C}\mathbf{\Omega}^\top$ are entry-wise positive. Then the function $f(\mathbf{B}) = \mathbf{1}_m^\top \{\mathbf{A} * \log(\mathbf{B}\mathbf{\Omega}) - \mathbf{C} * (\mathbf{B}\mathbf{\Omega})\} \mathbf{1}_n$ is non-decreasing under the multiplicative update

$$\mathbf{B} \leftarrow \mathbf{B} * \{[\mathbf{A} \oslash (\mathbf{B}\mathbf{\Omega})] \mathbf{\Omega}^\top\} \oslash (\mathbf{C}\mathbf{\Omega}^\top). \quad (17)$$

Proof. First, write the function f entrywise as

$$f(\mathbf{B}) = \sum_{ij} a_{ij} \log \left(\sum_r b_{ir} \omega_{rj} \right) - c_{ij} \sum_r b_{ir} \omega_{rj},$$

and the proposed update rule as

$$b_{ir} \leftarrow \frac{\bar{b}_{ir}}{\sum_j \omega_{rj} c_{ij}} \left(\sum_j \omega_{rj} \frac{a_{ij}}{\sum_r \bar{b}_{ir} \omega_{rj}} \right) \quad (18)$$

Now, let $\alpha_{rij} = \frac{\bar{b}_{ir}\omega_{rj}}{\sum_r \bar{b}_{ir}\omega_{rj}}$ and consider the function

$$g(\mathbf{B}, \bar{\mathbf{B}}) = \sum_{ijr} \left[\alpha_{rij} a_{ij} \log \left(\frac{b_{ir}\omega_{rj}}{\alpha_{rij}} \right) - c_{ij} b_{ir}\omega_{rj} \right].$$

Note, first, that

$$\begin{aligned} g(\mathbf{B}, \mathbf{B}) &= \sum_{ijr} \left[\alpha_{rij} a_{ij} \log \left(\frac{b_{ir}\omega_{rj}}{\alpha_{rij}} \right) - c_{ij} b_{ir}\omega_{rj} \right] \\ &= \sum_{ij} \frac{a_{ij}}{\sum_r b_{ir}\omega_{rj}} \left[\sum_r b_{ir}\omega_{rj} \log \left(\sum_r b_{ir}\omega_{rj} \right) \right] - \sum_{ij} c_{ij} \sum_r b_{ir}\omega_{rj} \\ &= \sum_{ij} a_{ij} \log \left(\sum_r b_{ir}\omega_{rj} \right) - \sum_{ij} c_{ij} \sum_r b_{ir}\omega_{rj} = f(\mathbf{B}). \end{aligned}$$

Then, consider that since $\alpha_{rij} \geq 0 \forall i, j, r$ and $\sum_r \alpha_{rij} = 1 \forall i, j$, the concavity of log gives us $\log(\sum_r b_{ir}\omega_{rj}) \geq \sum_r \alpha_{rij} \log \left(\frac{b_{ir}\omega_{rj}}{\alpha_{rij}} \right)$. So, $g(\mathbf{B}, \bar{\mathbf{B}}) \leq f(\mathbf{B})$. This, along with $g(\mathbf{B}, \mathbf{B}) = f(\mathbf{B})$, means $g(\mathbf{B}, \bar{\mathbf{B}})$ minorizes $f(\mathbf{B})$.

So, we consider the basic MM iteration $\mathbf{B}_{k+1} = \arg \max_{\mathcal{B}} g(\mathbf{B}, \mathbf{B}_k)$, which will be non-decreasing with respect to f since $f(\mathbf{B}_{k+1}) \geq g(\mathbf{B}_{k+1}, \mathbf{B}_k) \geq g(\mathbf{B}_k, \mathbf{B}_k) = f(\mathbf{B}_k)$. The first derivative of $g(\mathbf{B}, \bar{\mathbf{B}})$ is

$$\frac{dg(\mathbf{B}, \bar{\mathbf{B}})}{db_{ir}} = b_{ir}^{-1} \sum_j (\alpha_{rij} a_{ij}) - \sum_j (\omega_{rj} c_{ij}).$$

Setting the above to zero leads to $b_{ir} = \sum_j (\alpha_{rij} a_{ij}) / \sum_j (\omega_{rj} c_{ij})$, matching the entrywise version of our update rule (Equation 18). □

Theorem 2. Consider the scalar $\delta > 0$. The likelihoods $l_\delta(\mathcal{S}|\mathcal{X}, \mathcal{B})$ (Equation 15) and $l(\mathcal{S}|\mathcal{X}, \mathcal{B})$ (Equation 8) are related as

$$l_\delta(\mathcal{S}|\mathcal{X}, \mathcal{B}) = l(\mathcal{S}|\mathcal{X}, \mathcal{B}) - \frac{\delta}{2} \|\mathcal{S}\|_1.$$

Proof.

$$\begin{aligned} l_\delta(\mathcal{S}|\mathcal{X}, \mathcal{B}) &:= \frac{1}{2} \mathbf{1}^\top (\log(\mathbf{S}_d \mathbf{\Omega}_d) - (\mathcal{R}_{(d)} + \delta) * \mathbf{S}_d \mathbf{\Omega}_d) \mathbf{1}. \\ &= \frac{1}{2} \mathbf{1}^\top ((\log(\mathbf{S}_d \mathbf{\Omega}_d) - \mathcal{R}_{(d)} * \mathbf{S}_d \mathbf{\Omega}_d - \delta \mathbf{S}_d \mathbf{\Omega}_d)) \mathbf{1} \\ &= l(\mathcal{S}|\mathcal{X}, \mathcal{B}) - \frac{\delta}{2} \sum_{i,j} (\mathbf{S}_d \mathbf{\Omega}_d)_{i,j} = l^{(d)}(\mathcal{S}) - \frac{\delta}{2} \|\mathcal{S}\|_1. \end{aligned}$$

The final equality holds since all elements of $\mathcal{S}$ are positive by definition. □

F Computational Complexity

In this section, we provide the derivation of the time complexity of Algorithm 3 given in Equation 12, and provide the space complexity as well. Lower-order costs associated with

column normalization, storing scaling vectors, and convergence checks are omitted. Recall $N_{\times} = \prod_{d=1}^D N_d$, $N_+ = \sum_{d=1}^D N_d$, and $N_{-d} = \prod_{c \neq d} N_c = N_{\times}/N_d$.

First, we derive the cost for Algorithm 1. For mode d , the update is performed row-wise. The Khatri–Rao design matrix has size $(\mathbf{\Pi}_d)^{\top} \in \mathbb{R}^{N_{-d} \times R_{\mathcal{B}}}$. For each row $k = 1, \dots, N_d$, the update solving the weighted least-squares problem (Equation (7)) by a normal-equation, QR, SVD, or pseudoinverse-based calculation requires, to leading order, $O(N_{-d}R_{\mathcal{B}}^2 + R_{\mathcal{B}}^3)$ operations for each row. Since there are N_d rows in mode d , the cost for updating $\mathbf{B}_d$ is

$$O(N_d N_{-d} R_{\mathcal{B}}^2 + N_d R_{\mathcal{B}}^3) = O(N_{\times} R_{\mathcal{B}}^2 + N_d R_{\mathcal{B}}^3).$$

Summing over $d = 1, \dots, D$, one full sweep of Algorithm 1 costs $O(DN_{\times} R_{\mathcal{B}}^2 + N_+ R_{\mathcal{B}}^3)$. If $R \ll N_{-d} \forall d \in [D]$, typical for CP decomposition, this reduces to $O(DN_{\times} R_{\mathcal{B}}^2)$, the cost reported in Section 2.

For reference, a standard dense CP-ALS sweep with rank R has cost $O(DN_{\times} R + N_+ R^2 + DR^3)$, though we only reported $O(DN_{\times} R)$ in Section 2, since the MTTKRP (Matricized Tensor times Khatri Rao Product) cost will dominate the Gram and solve terms whenever

$$R \ll \frac{DN_{\times}}{N_+} \quad \text{and} \quad R^2 \ll N_{\times},$$

which is almost always the case for CP decomposition. The computational advantage of CP-ALS comes from the Khatri–Rao Gram identity

$$\mathbf{\Pi}_d \mathbf{\Pi}_d^{\top} = \underset{c \neq d}{*} (\mathbf{B}_c^{\top} \mathbf{B}_c),$$

which reduces the Gram calculation from the large $N_{-d} \times R$ Khatri–Rao matrix $(\mathbf{\Pi}_d)^{\top}$ to an $R \times R$ matrix. In contrast, the Gram matrix blocks $(\mathbf{G}_d) = \mathbf{\Pi}_d \text{diag}((\mathbf{S}_d)_{i_d,:}) \mathbf{\Pi}_d^{\top}$ used for Algorithm 1 depend $i_d \in [N_d]$, and do not form one shared $R_{\mathcal{B}} \times R_{\mathcal{B}}$ system for the entire mode update.

However, as pointed out in Section 2, when the weights $\mathcal{S}$ are low-rank, as assumed in this work, the HCP Gram matrices can be assembled more efficiently. Now, forming

$$\mathbf{G}_d = \sum_{r=1}^{R_{\mathcal{S}}} (\lambda_{\mathcal{S}})_r \text{diag}((\mathbf{S}_d)_{:,r}) \otimes \left(\underset{c \neq d}{*} \mathbf{H}_{c,r} \right),$$

where

$$\mathbf{H}_{c,r} := \mathbf{B}_c^{\top} \text{diag}((\mathbf{S}_c)_{:,r}) \mathbf{B}_c \in \mathbb{R}^{R_{\mathcal{B}} \times R_{\mathcal{B}}}, \quad r \in [R_{\mathcal{S}}], c \in [D]$$

costs only $O(N_+ R_{\mathcal{B}}^2 R_{\mathcal{S}})$, rather than $O(DN_{\times} R_{\mathcal{B}}^2)$. So, whenever

$$R_{\mathcal{B}} R_{\mathcal{S}} \ll \frac{DN_{\times}}{N_+} \quad \text{and} \quad R_{\mathcal{B}}^2 \ll \frac{DN_{\times}}{N_+}, \quad (19)$$

the $O(DN_{\times} R_{\mathcal{B}})$ MTTKRP in the normal equations (Equation (7)) becomes the dominate cost. So, for the parametrization of the precision we consider in this paper, with reasonably small ranks, Algorithm 1 has the same computational complexity as CP-ALS.

Next, let us consider Algorithm 2. The dense matrix products involving $\mathbf{\Omega}_d$ each cost $O(N_d N_{-d} R_{\mathcal{S}}) = O(N_{\times} R_{\mathcal{S}})$. Thus, a full sweep over all modes, with i_{in} inner multiplicative updates, costs $O(D i_{\text{in}} N_{\times} R_{\mathcal{S}})$.

For partially heteroskedastic precision, define $N_{\times,P} = \prod_{d=1}^P N_d$, where $P < D$ is the number of heteroskedastic modes (like in Section 4.2, we assume the tensor has been permuted such that heteroskedastic modes are the first P modes). The leading cost of a precision sweep then becomes

$$O(i_{\text{in}} P N_{\times,P} R_{\mathcal{S}}) < O(i_{\text{in}} D N_{\times} R_{\mathcal{S}}).$$

This reduction can be substantial when $N_{\times,P} \ll N_{\times}$.

In addition, many practical applications will only require $R_S = 1$. As we observed in Section 4.1, this makes the factor updates for Algorithm 2 deterministic, reducing the complexity from $O(Di_{\text{in}}N_{\times}R_S)$ to $O(DN_{\times}R_S)$.

Combining the mean and precision updates, a single sweep for each subroutine of Algorithm 3 has computational complexity

$$O(DN_{\times}R_B + i_{\text{in}}DN_{\times}R_S),$$

where we are assuming the ranks satisfy Equation (19). For the full alternating procedure, run for i_{tot} outer iterations, each with i_B sweeps Algorithm 1 and i_S sweeps Algorithm 2, the total time complexity is

$$O(i_{\text{tot}}[i_B DN_{\times}R_B + i_S i_{\text{in}} DN_{\times}R_S]).$$

Since i_{in} will be very small, R_S is often smaller than R_B , and we commonly leverage partial heteroskedasticity, we say that a single sweep for Algorithm 3 has roughly equal cost as a sweep for CP-ALS. Roughly, this means Algorithm 3 only has the cost of fitting CP-ALS $2i_{\text{tot}}$ times, making our method highly competitive from a time complexity standpoint.

Finally, the overall space complexity is

$$O(N_{\times} + N_+R_B + N_+R_S),$$

where $O(N_{\times})$ is from storing the tensors, and $O(N_+R_B + N_+R_S)$ for storing the factor matrices. For partially heteroskedastic precision models, the precision-factor storage becomes $O(R_S \sum_{d=1}^p N_d)$.

G Model Selection

In this section we give our recommendations for how to choose the user specified parameters in Algorithms 1, 2, and 3.

In the examples in the main manuscript, we have treated the ranks R_{mean} and R_{var} as true and known, or tailored the ranks to the application. In Appendix D we demonstrated that a misspecification of ranks will not drag HCP’s performance below that of standard CP decomposition, and the performance will be fairly robust to overestimates of the fitted ranks. However, a data-driven mechanism for model selection is often desirable. In this section, we will provide options for rank selection, as well as provide recommendations for choosing the other user-specified inputs.

The simplest evaluation metrics would be to simply use RSS $\|\mathcal{X} - \hat{\mathcal{B}}\|_F^2$ or Relative Error $\frac{\|\mathcal{X} - \hat{\mathcal{B}}\|_F}{\|\mathcal{X}\|_F}$, or their $\hat{S}$ weighted counterparts. However, these typically decrease with ranks R_B and R_S , so they are not a great measures of “goodness-of-fit” for model selection. In their textbook on tensor decompositions, Ballard and Kolda [2025] advocate for an elbow-style approach of selecting smallest R for which $R - 1$ only has minor improvement in RSS or relative error, which we can adapt to HCP by comparing the present (weighted, if desired) relative error of $R_{\text{mean}}, R_{\text{var}}$ to both $R_{\text{mean}} + 1, R_{\text{var}}$ and $R_{\text{mean}}, R_{\text{var}} + 1$. CONCORDIA Bro and Kiers [2003] or NORMO Fernandes et al. [2020] propose “appropriateness” measures for goodness of fit, which rely on structural interpretations of the factor matrices.

Since we take a likelihood-based approach, we can leverage Akaike information criterion (AIC) or Bayesian information criterion (BIC) to select the model. BIC, for example, would simply be

$$-2\ell(\mathcal{B}, \mathcal{S}|\mathcal{X}) + \log(N_{\times})[(R_B + R_S)N_+ - (R_B + R_S)(D - 1)],$$

where parameter counts take away $D - 1$ degrees of freedom per component due to the scaling non-identifiability discussed in Section 2. Many other likelihood-based tensor methods leverage

AIC or BIC similarly, see Wang et al. [2023] for AIC or Zhang et al. [2019] and Ke et al. [2023] for BIC.

We can also adapt a type of cross validation by masking a proportion of our entries, and solving for the low-rank estimate as a tensor completion problem. This is not typically used for standard CP decomposition, since the tensor completion problem is solved differently than the original decomposition (often with imputation (Tomasi and Bro [2005]), or with weights (Acar et al. [2011b]) like we do in Section 2), but likelihood-based methods lend themselves to tensor completion immediately [Hong et al., 2020, Hood and Schein, 2026]. Suppose that we observe $\mathcal{X}$ partially, and consider an index tensor $\mathcal{T}$ such that $\mathcal{T}_i = 1$ if the entry $\mathcal{X}_i$ is observed, and zero otherwise. Our new loss function is:

$$\ell(\mathcal{B}, \mathcal{S}) = -\frac{1}{2} \sum_j \mathcal{T}_j (-\log(\mathcal{S}_j) + \mathcal{S}_j(\mathcal{X}_j - \mathcal{B}_j)^2),$$

which can be solved with nearly the same alternating procedure we use for the complete data (we provide exact details for the tensor completion implementation in Appendix H). So, we can assess the fit of our low-rank model by analyzing relative errors in the completion problem as we mask different entries in our data.

The convergence criterion, initialization, and maximum iterations are also user specified parameters for Algorithms 1, 2, and 3. Since CP decompositions are not unique, we recommend using stationarity of the recovered tensor or of the likelihood, rather than of the factors, as convergence criterion. That being said, the algorithm will typically converge quickly, so for all experiments in this paper we simply use a small number of iterations for $i_{\mathcal{B}}$, $i_{\mathcal{S}}$, and i_{tot} . For i_{in} , one should use an especially small number of iterations because it is more beneficial to move on to the next factor with an improved estimate than it is to solve for the current factor perfectly. In our implementation we used $i_{\mathcal{B}} = 5$, $i_{\mathcal{S}} = 5$, $i_{\text{in}} = 5$, and $i_{\text{tot}} = 20$.

For initialization, we mentioned in Section 4 that we use standard CP decomposition to initialize the mean factors, and then run Algorithm 2 with a larger number of iterations to initialize the precision factors (we used 50). We only note here that we recommend using multiple seeds and selecting the result with the largest final likelihood, as ALS methods can occasionally get stuck in poor local optima, particularly in settings with missing data. For our EEG experiment in Section 6, we used the best of 5 initializations for both HCP and CP.

Finally, we note that the choice of homoskedastic modes or residual constraints in Sections 4.2 and 4.3 are not viewed as parameters that need to be selected in practice the same way as the ranks. We would expect directions of homoskedastic variance to be chosen by the context or subject matter expertise prior to rank selection. If there is no contextual reason to consider partially heteroskedastic precision, we recommend simply defaulting to full heteroskedasticity, rather than trying to perform model selection over choices of homoskedastic modes and ranks simultaneously. For the lower bound on the residuals δ , we implement as the default value $\delta = \sqrt{\text{Machine-Precision}}$, and only recommend increasing this if the degeneracy occurs. This is very easy to spot in practice and can be done simultaneously with rank selection; if a single entry of $\mathcal{S}$ is many orders of magnitude larger than the others, then δ can be increased a few orders of magnitude. Like we see in Section 5.3, the size of δ typically has little impact on the performance of HCP.

H Tensor completion

Suppose that we observe $\mathcal{X}$ partially, and consider an index tensor $\mathcal{T}$ such that $\mathcal{T}_i = 1$ if the entry $\mathcal{X}_i$ is observed, and zero otherwise. Our new loss function is:

$$\ell(\mathcal{B}, \mathcal{S}) = -\frac{1}{2} \sum_j \mathcal{T}_j (-\log(\mathcal{S}_j) + \mathcal{S}_j(\mathcal{X}_j - \mathcal{B}_j)^2)$$

For fixed $\mathcal{S}$, we simply use Algorithm 1 with weights $\mathcal{T} * \mathcal{S}$. Note, though, that we should no longer expect the efficiency improvement from low-rank $\mathcal{S}$, though, since, unless $\mathcal{T}$ is appropriately structured, $\mathcal{T} * \mathcal{S}$ will not be low-rank.

For fixed $\mathcal{B}$, we follow the same MM argument as in Section 3 to get the following mode- d precision-factor update:

$$\mathbf{S}_d \leftarrow \mathbf{S}_d * ([\mathcal{T}_{(d)} \oslash (\mathbf{S}_d \mathbf{\Pi}_d)] \mathbf{\Pi}_d^\top) \oslash ((\mathcal{T}_{(d)} * \mathcal{R}_{(d)}) \mathbf{\Pi}_d^\top). \quad (20)$$

Rank-one precision, partially heteroskedastic precision, and constrained residuals are handled by following the steps described in Sections 4.1, 4.2, and 4.3 to the modified likelihood. In the partially heteroskedastic case, we note the denominator in Equation (14) is the number of observed entries, instead of the size of the homoskedastic modes. This distinction is important under missingness, since different retained entries may summarize different numbers of observed tensor entries. These cases are included in the tensor completion implementation, but we do not provide full algorithmic details here.

We note that the final Hadamard divisor in Equation 20 is far more likely to have an entry of 0 than its counterpart in Equation 11, making including a constraint as in Section 4.3 even more necessary. This happens when all observed entries for a row of matricized squared residuals are 0, becoming more likely when fewer entries are observed.

Finally, if an entire row of $\mathcal{X}_{(d)}$ is unobserved, then the corresponding factor row is unidentifiable from the observed likelihood, and this procedure cannot determine it. We also expect a rank condition on the columns of $\mathbf{\Pi}_d$ selected by the observed entries in each row of $\mathcal{X}_{(d)}$, analogous to that in CP-APR Chi and Kolda [2012], whose condition implies at least R observed entries in each row of $\mathcal{X}_{(d)}$, for all $d \in [D]$. We leave a full analysis of the tensor-completion properties of HCP to future work.

I Generalized CP Decomposition Implementation Details

The experiments in Sections 5.1 and 5.2 compare HCP with Generalized CP (GCP) [Hong et al., 2020]. We use this section to provide details on our GCP implementation.

For a data tensor $\mathcal{X}$, GCP estimates a low-rank tensor $\hat{\mathcal{B}} = \llbracket \hat{\boldsymbol{\lambda}}; \mathbf{B}_1, \dots, \mathbf{B}_D \rrbracket$ by solving

$$\min_{\hat{\mathcal{B}}} \sum_j f(\mathcal{X}_j, \hat{\mathcal{B}}_j),$$

where f is chosen to match a distributional model. The estimated low-rank tensor corresponds to the mean or natural parameter of the chosen loss.

All GCP competitors were fit using `pyttb` [Dunlavy et al., 2026] with an L-BFGS-B optimizer. We matched ranks to HCP’s $R_{\mathcal{B}}$.

The Gamma GCP used the built-in `pyttb` Gamma objective. For Poisson GCP, we did not use the default Poisson objective wrapper, which expects integer count data. Instead, we passed the Poisson loss and gradient handles directly. For the inverse-Gaussian GCP, we used the entrywise loss and gradient

$$f(x, m) = \frac{(x - m)^2}{2xm^2}, \quad \frac{\partial f}{\partial m} = \frac{m - x}{m^3}.$$

The Gamma and inverse-Gaussian objectives require positive observations, and the corresponding synthetic datasets were generated to satisfy this constraint.

For the butterfly experiment, the Gaussian-noise observations can be negative, while the Poisson GCP objective is defined for nonnegative data. So, for each noisy tensor $\mathcal{X}$, we set $c = \max\{0, -\min_j \mathcal{X}_j + 10^{-8}\}$, fit Poisson GCP to $\mathcal{X} + c$, and then shifted the fitted tensor back to report the relative error.

J EEG Experiment Details

This appendix provides additional details on the preprocessing steps used to construct the EEG tensor analyzed in Section 6, as well as some additional results.

We use the publicly available EEG Motor Movement/Imagery Dataset from PhysioNet [Goldberger et al., 2000, Schalk et al., 2009], collected using the BCI2000 system [Schalk et al., 2004]. The full dataset contains 64-channel EEG recordings from 109 subjects, with each subject completing 14 experimental runs. The EEG signals were recorded from 64 electrodes placed according to the international 10-10 system, excluding electrodes Nz, F9, F10, FT9, FT10, A1, A2, TP9, TP10, P9, and P10. Thus, throughout Section 6, a “channel” refers to one of these electrode locations on the scalp. We generated the scalp maps in Figure 7 from MNE-Python [Gramfort et al., 2014].

In our analysis, we use a single subject performing task 1, where a target appears on either the left or the right side of the screen, and the subject opens and closes the corresponding fist until the target disappears. This corresponds to runs 3, 7, and 11 in the PhysioNet protocol. The event annotations $T1$ and $T2$ correspond to left- and right-fist movement onsets, respectively, while $T0$ corresponds to rest. We only retain the task-related $T1$ and $T2$ epochs. Each epoch begins at the annotated movement onset and extends from 0 to 4 seconds relative to that onset. Since the PhysioNet EEG Motor Movement/Imagery data are sampled at 160 Hz, each un-decimated epoch contains 641 time samples, including both endpoints.

The raw EEG recordings are loaded from the EDF files, the EEGBCI channel names are standardized, and a standard 10-05 montage is assigned using MNE-Python. We retain only EEG channels and apply a 1–40 Hz bandpass filter prior to the time-frequency transform.

For each epoch and channel, we compute a complex Morlet wavelet time-frequency transform. Let $x_{i,k}(t)$ denote the EEG signal from channel i during epoch k . For each frequency f , the Morlet transform produces complex coefficients of the form $W_{i,f,t,k} \in \mathbb{C}$, indexed by channel i , frequency f , time t , and epoch k . In our implementation, we use the frequency grid $f \in \{4, 6, 8, \dots, 40\}$ Hz, giving 19 frequencies in total. The number of Morlet cycles is chosen to depend on frequency according to $n_{\text{cycles}}(f) = \frac{f}{2}$. Thus, for example, the 4 Hz wavelet uses 2 cycles, the 10 Hz wavelet uses 5 cycles, and the 40 Hz wavelet uses 20 cycles. The time-frequency coefficients are decimated by a factor of 2, so the effective time grid has spacing $2/160 = 0.0125$ seconds. Over the 0–4 second epoch window, this yields 321 time samples. We convert the complex-valued coefficients to real-valued features by taking their magnitudes, leaving, for each channel and epoch, $19 \times 321 = 6099$ real-valued coefficients.

The resulting array initially has dimensions

$$\text{epoch} \times \text{channel} \times \text{frequency} \times \text{time}.$$

For each channel and epoch, the frequency-by-time matrix of wavelet magnitudes is vectorized into a single time-frequency feature vector. Stacking these vectors yields a third-order tensor

$$\mathcal{X} \in \mathbb{R}^{64 \times 6099 \times 45},$$

whose modes correspond to channel, vectorized time-frequency coefficient, and measurement/epoch. The 45 measurements correspond to the $T1$ and $T2$ task epochs extracted from the three task-1 runs.

Finally, as in Acar et al. [2011b], we center the tensor across the channel mode. Specifically, for each time-frequency coefficient j and epoch k , we subtract the mean over channels:

$$\tilde{X}_{i,j,k} = X_{i,j,k} - \frac{1}{64} \sum_{i'=1}^{64} X_{i',j,k},$$

and rescale the tensor entries by 10^6 , so that the final tensor used in the factorization is expressed on a microvolt scale.

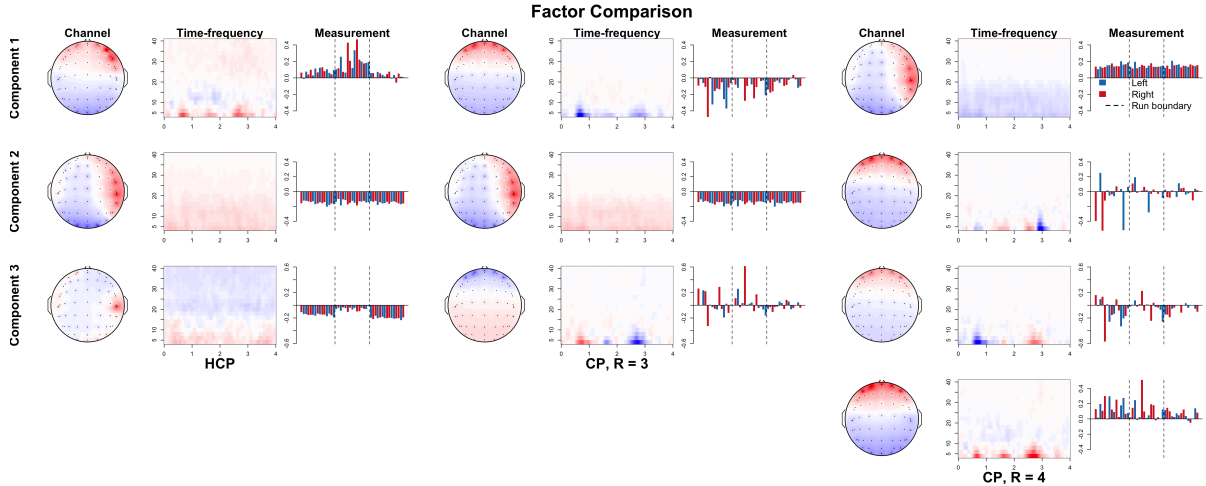


Figure 16: Visualization of the factor matrices recovered from the EEG tensor by HCP and CP with rows corresponding to components. Within each method, columns show the channel/scalp loadings, time-frequency loadings, and measurement loadings. Dashed vertical lines in measurement columns indicate recording-run boundaries.

Our preprocessing follows the structure used by Acar et al. [2011b]: complex wavelet coefficients are converted to real values using magnitudes, the data are arranged as a channel-by-time-frequency-by-measurement tensor, and the tensor is centered across the channel mode. However, their EEG example uses a proprietary dataset and reports a time-frequency vector of length 4392, whereas our PhysioNet preprocessing uses 19 frequencies and 321 time samples, yielding 6099 time-frequency coefficients.

Finally, we present an additional result, where we fit CP decomposition with $R = 4$. Since HCP requires extra parameters to fit the precision tensor, the total number of parameters with $R_B = 3$ and $R_S = 1$ is nearly² the same number of parameters as CP with $R = 4$, making comparisons to both CP with $R = 3$ and $R = 4$ logical. The recovered factors, alongside the original factors with $R = 3$ from Figure 7, are presented in Figure 16. As we can see, the additional component does not allow CP to capture the localized channel signal that HCP captures in the third component. Instead, CP uses this extra component to further model the low-frequency pattern that is most likely variance, rather than true signal.

K Computing Infrastructure & Software

Computing Infrastructure. The experiments in Sections 5.1, 6, and C were run on a 16-inch MacBook Pro with 2.4 GHz 8-Core Intel Core i9 CPU, 32 GB of RAM, and no dedicated GPU.

The experiments in Sections 5.3, 5.2, and D were run on the single node of an HPC cluster equipped with dual 14-core Intel Xeon processors (28 physical cores per node). We used 1 node, 1 task, and 4 CPUs per task, and set OpenMP/BLAS threading to match:

Software. EEG preprocessing was performed in Python using MNE-Python [Gramfort et al., 2014], including the EEGBCI data-loading utilities and the Morlet time-frequency transform. Tensor computations were performed in R using `rTensor` [Li et al., 2018]. Experiments involving generalized CP tensor factorization used the `pyttb` implementation [Hong et al., 2020, Dunlavy et al., 2026].

²Since $\mathcal{S}$ is partially heteroskedastic, it has $64 + 6099$ parameters, while an additional rank for CP adds $64 + 6099 + 45$ parameters